

THE STRESS ANALYSIS OF CRACKS HANDBOOK

THIRD EDITION

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DEDICATION

On 9 October 1998, Dr. George R. Irwin passed away. We, the surviving authors of this book, have lost a great friend and colleague, who deservingly has been called “the Father of Fracture Mechanics.” He was our inspiration in continuing to develop new items for the second and third editions of this book. Therefore, we feel compelled to dedicate our effort on this third edition to him, as well as our effort on previous editions. We are proud of the privilege of having worked closely with him.

Paul C. Paris and Hiroshi Tada, May 2000

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LIST OF PRINCIPAL SYMBOLS

Only the principal symbols are listed here. So many symbols are needed that the notation is not necessarily consistent throughout the book. However, most symbols are clearly defined within each solution page and those not included in this list will be readily identified.

In many two-dimensional configurations, forces and moment (P , Q , etc., and M) implicitly designate forces and moment unit thickness. The definition will most often be obvious from the context; otherwise, confirm the definition so that the stress intensity factor, K , yielded has the dimension: force/(length)^{3/2} or stress · (length)^{1/2}.

I, II, III (subscripts)	Designations for Mode I, Mode II, and Mode III, respectively
A	Area of crack surface Area of crack opening A crack tip (2-D) or a point on crack front (3-D)
A_n	Coefficients for series expansion of $Z(z)$
A_{ij}	Elastic constants for anisotropic solid
a	Half length of crack in plate or shell Depth of edge crack Length or half length of net ligament (2-D) Radius of circular crack or circular net ligament Semi-major axis of elliptical crack or elliptical net ligament
a_{ij}	Elastic constants for anisotropic solid
B	A crack tip (2-D) or a point on crack front (3-D) Width of crack surface for rod or beam
$B(B_I, B_{II}, B_{III})$	Strength of Bueckner-type crack-tip singularity

b	Width or half width of strip Depth of bend specimen Radius of cracked disc or circular hole Radius of round bar with circular crack Semi-minor axis of elliptical crack or elliptical net ligament Semi-major axis of elliptical hole or depth of semi-elliptical notch Half width of rectangular hole Coordinate or length-defining position of concentrated or distributed load Half length of second crack (collinear cracks) or of vertical crack (cruciform crack)
C	Elastic compliance Elastic constant: 1 for plane stress, $1/\sqrt{1-\nu^2}$ for plane strain
$C_{ij}(a)$	Elastic compliance of cracked body at i th load point by j th load
c	Coordinate or length defining position of concentrated or distributed load Semi-major or minor axis of elliptical hole or depth of semi-elliptical notch Radial width of annular crack Model crack size for strip yield analysis (actual crack size + plastic zone size) Model net ligament size for strip yield analysis (actual ligament size - plastic zone size) Coordinate of zero crossing of residual stress distribution
D	Diameter of (cracked) disc
$D(x/a), D(\gamma)$, etc.	Configuration functions for crack opening displacement
d	Semi-minor axis of elliptical hole or half width of semi-elliptical notch Distance between center of crack and center of circular hole
E	Young's modulus
E'	Elastic constant = $4 \alpha G : E$ for plane stress, $E/(1-\nu^2)$ for plane strain
$E(k)$	Complete elliptic integral of the second kind
$E(\varphi, k)$	Elliptic integral of the second kind
e	Eccentricity of crack in strip Base of natural logarithm
F	Concentrated force

$F(a/b), F(\theta), F(\lambda)$ $F_I(a/b), F_2(a/b)$ $F_I(a/b), F_{II}(a/b)$ $F_{III}(a/b), \text{ etc.}$	Configuration correction factors for stress intensity factor
$F(\varphi, k)$	Elliptic integral of the first kind
$f_m(s, a)$	Weight function determined from m th loading system
G	Shear modulus
$G(a/b), \text{ etc.}$	Alternate forms of configuration correction factors for stress-intensity factor
$G(\theta), G(\lambda), \text{ etc.}$	Configuration functions for crack opening area for shells
$\mathcal{G}(\mathcal{G}_I, \mathcal{G}_{II}, \mathcal{G}_{III})$	Crack extension force
H	Distance between parallel cracks
$H(a/b), \text{ etc.}$	Configuration functions for crack opening displacement
h	Half depth of beam Height or half height of strip with crack vertical to side edges Half width of strip with crack parallel to side edges Half distance between parallel cracks Half width of rectangular hole Half thickness of uniform wedge Distance from free surface to crack parallel to it in semi-infinite plate
$I, I_1, I_2, \text{ etc.}$	Moments of inertia of cross section of rods or beams
$I(\theta)$	Configuration function for crack opening area for cylindrical shell
$\text{Im}()$	Imaginary part of ()
J	J -Integral
$K(K_I, K_{II}, K_{III})$	Stress intensity factor
$K_{I \pm a}, K_{IA}, \text{ etc.}$	K_I at $x = \pm a$, K_I at A , etc.
K_t	Stress concentration factor
$K(k)$	Complete elliptic integral of the first kind
k	Modulus of elliptic integral
k'	Complementary modulus of elliptic integral

$L, L', L_1, L_2, \text{ etc.}$	Dimensions defined in solution pages
ℓ	Length of normal for ellipse Total length of crack and hole (or notch) Length of plastic zone for strip yield model Length of crack surface contact
ℓ_1, ℓ_2 ℓ_a, ℓ_b $\ell_\alpha, \ell_\beta, \text{ etc.}$	Dimensions defined in solution pages
M	Bending moment In-plane moment per unit thickness
m	Slope of tapered double cantilever specimen
$()_{\max}$	Maximum value of ()
$()_{\min}$	Minimum value of ()
n	Number of radial cracks for star-shaped crack
o	Origin of coordinate system
P	Concentrated load Concentrated load per unit thickness (2-D)
P_i	i th applied load
p	Distributed load (stress) Internal pressure in shell
$\bar{p}$	Line force (force/length)
Q	Concentrated load Concentrated load per unit thickness (2-D)
q	Distributed load (stress)
R	Concentrated load Radius of circular arc crack Radius of circular hole Radius of cylindrical or spherical shell
$Re()$	Real part of ()
r	First (radial) coordinate of polar $[r, \theta, (z)]$ coordinate system
r_i, r_o	Inner and outer radius, respectively, of thick-walled cylinder

r_p, r_Y	Plastic zone size and size index, respectively ($r_p = 2r_Y$)
$S(a/b)$, etc.	Configuration functions for crack opening area or rotation
s	Span between supports of bend specimen Distance from crack plane to point of concentrated load application Normalized half vertex angle of cracked wedge Dimensionless parameter characterizing geometry of cracked body, e.g., $s = a/(a + b)$
T	Concentrated load per unit thickness (Mode III) Twisting moment
$T(s)$	Distributed traction over surface s
t	Distributed load (stress) Thickness of shell
U	Elastic energy density
U_T	Total strain energy in cracked body
$U(a/b)$, $U_1(a/b)$ $U_2(a/b)$, etc.	Configuration functions for displacement
u	x -direction displacement in (x, y, z) system
u_i	Load point displacement of i th load P_i
u_r, u_θ	Displacement components in (r, θ) system
V	Volume of crack (3-D)
$V(a/b)$, $V_1(a/b)$ $V_2(a/b)$, etc.	Configuration functions for displacement
$V_1(a)$, $V_1(a_0)$ $V_2(a)$, $V_2(a_0)$	Electric potentials
v	y -direction displacement in (x, y, z) system
$v(x, 0)$, $v(r, 0)$	Half crack opening displacements
v_s, v , $v(0, y)$ $v(0, s)$, $v(0, 0, s)$, etc.	Displacements perpendicular to crack plane
W	Width of plate (strip) Interval of periodically repeated collinear cracks
$W(a/b)$, $W_1(a/b)$ $W_2(a/b)$, etc.	Configuration functions for displacement (Mode III)

w	z -direction displacement in (x, y, z) system
X	x -coordinate of point A(X, Y) on circular crack front
x	First coordinate of (x, y, z) Cartesian coordinate system Real part of complex variable $z = x + iy$
x_0	x -coordinate of point of concentrated load application
Y	y -coordinate of point A(X, Y) on circular crack front
y	Second coordinate of (x, y, z) Cartesian coordinate system Imaginary part of complex variable $z = x + iy$
y_0	y -coordinate of point of concentrated load application
$Z(z)$ ($Z_I(z), Z_{II}(z), Z_{III}(z)$)	Westergaard stress function
$Z'(z), \bar{Z}(z), \bar{\bar{Z}}(z)$	$Z'(z) = dZ(z)/dz, Z(z) = d\bar{Z}(z)/dz, \bar{Z}(z) = d\bar{\bar{Z}}(z)/dz$
$Z^*(z)$	Complex conjugate of $Z(z)$ (used in Part I)
z	Third coordinate of (x, y, z) or (r, θ, z) coordinate system Complex variable $z = x + iy$
$\bar{z}, z^*$	Complex conjugate of z : $\bar{z} = x - iy$ (z^* is used in Part I)
z_0	Complex coordinate of point of concentrated load application ($z_0 = x_0 + iy_0$)
α	Elastic constant: $(1+\nu)/2$ for plane stress, $1/[2(1-\nu)]$ for plane strain ($= 1/\beta$) Direction of applied stress Direction of concentrated load (for cracked wedge)
β	Elastic constant ($= 1/\alpha$) Parameter characterizing relative rigidity of second plate (or material) ($= E_1 h_1 / Eh$)
$\Gamma(x)$	Gamma function
γ	Direction of applied stress Exponent specifying distributed load
$\gamma_{xy}, \gamma_{yz}, \gamma_{zx}$	Shear strain components in (x, y, z) coordinate system
$\gamma_{r\theta}, \gamma_{\theta z}, \gamma_{zr}$	Shear strain components in (r, θ, z) coordinate system
Δ	Thickness of uniform wedge

$\Delta (\Delta_I, \Delta_{II}, \Delta_{III})$	Load point displacement
Δ_i	Load point displacement at i th load (P_i)
Δ_{crack}	Additional displacement due to the presence of crack
$\Delta_{\text{no crack}}$	Displacement in the absence of crack
Δ_{total}	Total displacement ($= \Delta_{\text{no crack}} + \Delta_{\text{crack}}$)
$\delta (\delta_I, \delta_{II}, \delta_{III})$	Crack surface relative displacement
$\delta_0, \delta_b, \delta(x)$, etc.	Crack opening displacements at specific point on crack (Mode I)
ε	Shell parameter: $\varepsilon^2 = (t/R)\sqrt{12(1-\nu^2)}$
$\varepsilon_x, \varepsilon_y, \varepsilon_z$	Normal strain components in (x, y, z) coordinate system
$\varepsilon_r, \varepsilon_\theta, \varepsilon_z$	Normal strain components in (r, θ, z) coordinate system
ζ	Complex coordinate taken at crack tip
θ	Second coordinate (polar angle) in $[r, \theta, (z)]$ coordinate system Parametric angle defining point A on crack front of circular, elliptical, semi-circular (surface), or quarter-circular (corner) crack Half central angle contained by circular arc crack(s) (2-D) or circumferential crack in cylindrical shell Half vertex angle of infinite wedge with symmetrical edge crack Relative rotation at infinity
θ_{crack}	Additional rotation due to the presence of crack
$\theta_{\text{no crack}}$	Rotation in the absence of crack
θ_{total}	Total rotation ($= \theta_{\text{no crack}} + \theta_{\text{crack}}$)
κ	Elastic constant: $(3-\nu)/(1+\nu)$ for plane stress, $3-4\nu$ for plane strain ($= 2\beta-1$)
λ	Ratio of two systems of applied load Shell parameter ($= \theta/\sqrt{t/R} = a\sqrt{Rt}$)
ν	Poisson's ratio
$\Pi(\varphi, n, k)$	Elliptic integral of the third kind

ρ	Notch tip root radius Mass density
σ, σ'	Applied stress
σ_0	Maximum tensile stress in residual stress field
$\sigma_x, \sigma_y, \sigma_z$	Normal stress components in (x, y, z) coordinate system
$\sigma_r, \sigma_\theta, \sigma_z$	Normal stress components in (r, θ, z) coordinate system
σ_Y, σ_{YP}	Yield strength in tension
τ, τ_ℓ	Applied shear stress in Mode II and Mode III, respectively
τ_Y	Yield strength in shear
$\tau_{xy}, \tau_{yz}, \tau_{zx}$	Shear stress components in (x, y, z) coordinate system
$\tau_{r\theta}, \tau_{\theta z}, \tau_{zr}$	Shear stress components in (r, θ, z) coordinate system
Φ	Airy's stress function
$\Phi(a/b), \Phi(\theta),$ $\Phi_1(\theta), \Phi_2(\theta), \text{ etc.}$	Configuration functions for rotation
$\phi, \phi(z)$	Function of complex variable $z = x + iy$
ϕ	Relative rotation at infinity or kink angle ($= \phi_{\text{crack}}$) at cracked section
ϕ_{crack}	Additional rotation (or kink) due to the presence of crack
$\phi_{\text{no crack}}$	Rotation in the absence of crack
ϕ_{total}	Total relative rotation ($= \phi_{\text{no crack}} + \phi_{\text{crack}}$)
$\chi, \chi(z)$	Function of complex variable $z = x + iy$
ω	Angular velocity of rotating disc Angle defining direction of applied load
$\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial y_0}, \frac{\partial}{\partial s}$ $\frac{\partial^2}{\partial x^2}, \frac{\partial^2}{\partial y^2}, \frac{\partial^2}{\partial x \partial y}, \text{ etc.}$	Partial differential operators
∇^2	Harmonic operator ($= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ in (x, y) system)

FOREWORD

Fracture mechanics was introduced in the 1947-1952 period using the idea that onset of rapid crack extension occurred when the crack extension force became large enough to cause rapid joining of small openings near the leading edge of a crack. The “force” concept used was the rate of loss of stress field energy, G , per unit of new separation area. Unfortunately the usual training in stress analysis of engineers did not provide methods of estimating values of G . However, in the mid-1950s, use of a relatively simple method of crack-stress field analysis, introduced by Westergaard, permitted demonstration that the severity of the enclosing stress field, tending to cause crack extension, could be represented by a stress intensity factor, K .

In addition, values of the force, G , were related to K by the use of equation $G = K^2/E$; where E is Young’s modulus. This led to use of toughness measurements in terms of critical values of K necessary for rapid crack extension. This change of concept and nomenclature was of special importance to the understanding and practical use of fracture mechanics by engineers, and led immediately to general acceptance of fracture mechanics. Despite the sound theoretical basis for the force, G , engineers preferred a representation of critical conditions for crack extension in terms of principles of stress analysis with which they were familiar.

The introduction of the K concept was shown to be of special value for studies of fatigue cracking. It was shown that from calibration tests, it was possible to make estimates of the danger of crack growth by small initial cracks due to load fluxuations during periods of use in service. The use of K values for studies of fatigue cracking was followed by the use of K values for studies of corrosion cracking and corrosion fatigue.

In the use of fracture mechanics, estimates of K for potential or real cracks are commonly needed. For this purpose, Tada’s Handbook of K Values (renamed The Stress Analysis of Cracks Handbook) for cracks in various structural locations has been widely used. Previously available only in notebook form, this collection of K values has been reviewed and checked carefully.

G. R. Irwin, 1997

PREFACE — THIRD EDITION

The work on this handbook virtually began during the doctoral dissertation of Dr. Hiroshi Tada under the direction of Dr. George R. Irwin during the late 1960s at Lehigh University. In that dissertation, a modest number of new crack-tip stress intensity factor solutions were developed. Upon completion of his degree, Dr. Tada was employed by Del Research Corporation with the primary task of developing material for the Stress Analysis of Cracks Handbook. That led directly to the two earlier editions, with Fracture Proof Design Corporation providing the venue for much of the work on the latter of the two. This third edition has been produced with Dr. Paul C. Paris and Dr. Tada at Washington University, St. Louis, with modest cooperative effort from Dr. Irwin. It seems fitting that this long-term effort to develop such a handbook should finally be published hardbound by a leading engineering society, The American Society of Mechanical Engineers (ASME).

During the 30 years of development of this work, Dr. Tada has continuously devised new solutions, collected others and improved them, and developed fitting formulas and curves to present them in a convenient form for use by practitioners and researchers alike in the field of fracture mechanics. His coauthors herewith recognize his monumental effort in accomplishing that task. The text accompanying the solutions presented in this handbook was the joint task of the three individuals involved, each contributing several sections and editing others. In addition, we acknowledge that the original work related to three of the appendices included contributions from other coauthors: H. Ernst, R. McMeeking, and L. L. Loushin. The involvement of many other individuals through direct assistance, suggestions, corrections, and encouragement throughout the 30 years are also noted and appreciatively acknowledged.

There is a software disk (see pages 676–677) available to purchasers of the third edition which allows rapid numerical computation of some much-used stress intensity factor K formulas for commonly adopted test and crack configurations found in practice. We especially thank Drs. Dilip Dedhia and David Harris for their interest in this book. Incidentally, the new appendix on K values for plates subjected to pinching loads was in fact originally a topic raised by the dissertation of Dr. Harris.

This third edition also adds new appendix sections on the J-Integral, on displacements prescribed on crack surfaces, on plastic zone instability (explaining a potentially interceding “elastic” failure mechanism), on engineering estimates of stress intensity factors, and Mode III plasticity solutions, as well as about 30 new solution pages and modifications of many older solutions.

The objective of this and each edition has been to document all of the important methods and results of elastic stress analysis as may be applied to small-scale yielding fracture mechanics and beyond. The principles and methods are found in the initial text sections and in the many appendices provided. Numerical approaches such as finite element methods, boundary collocation methods, and so on, remain in such a high state of development that discussion of them has been deliberately omitted. However, we have attempted to include all of the relevant and lasting material on elastic analysis as it applies to fracture mechanics and related disciplines.

Hiroshi Tada, Paul C. Paris, George R. Irwin, September 1997

PREFACE — SECOND EDITION

Since the last modification of this handbook in 1975, many new results have been forthcoming that are appropriate to include herein. Over 100 new solutions and other material have been added. Some corrections and modifications for completeness of existing crack stress analysis solutions have also been included in the new edition.

The project of further developing this handbook is ongoing and we hope to offer additional results some time in the future.

We thank the many readers who have offered comments and corrections over the past 10+ years. Further suggestions are welcome.

Hiroshi Tada, Paul C. Paris, October 1985

PREFACE — FIRST EDITION

This text is intended to provide the user with a comprehensive source of formulas and stress analysis information on crack problems. The emphasis is on useful information for treatment of actual problems on crack propagation through fracture mechanics correlation parameters and current fracture criteria, such as K_I approaching K_{Ic} as a plane strain fracture criterion.

The information provided, however, is not limited to that used in current practice, but also embodies other fundamental stress analysis results. For example, where stress functions are known for the complete solution to a crack problem they are either listed or referenced; again, where they are known, functions are listed that may be readily converted to displacements, such as integrals of stress functions.

Each numerical solution and approximation method is accompanied by the author's estimate of the accuracy of the results or the method; moreover, source references are listed in all cases for those users who wish to explore further details.

The information presented is useful only to the degree that it can be understood and properly used. For this reason, descriptive sections of text material are included (a) to define the meaning of the information presented, (b) to indicate and illustrate its conversion to other forms, and (c) to develop methods of applying it to actual cases or problems.

In addition, there are sections of text devoted to (a) the theory and useful methods of compliance calibration analysis; (b) weight function analysis for handling certain cases of arbitrary loading; (c) orthotropic, anisotropic, and dynamic effects; and (d) plasticity analysis of crack problems, especially a discussion of the J -Integral methods. Other implications of crack stress analysis (e.g., stress concentrations and notch field equations) and related results (e.g., electric fields in plates with cracks for electrical potential calibration) are given where available.

Obviously, we intended not to limit the material presented to idealized stress analysis results alone, but rather to expand those results where usefulness will be enhanced. For that reason any suggestions by the reader for future additions are welcome.

ACKNOWLEDGMENTS TO THE FIRST EDITION

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Hiroshi Tada, Paul C. Paris, George R. Irwin, June 1973

INTRODUCTORY INFORMATION

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INTRODUCTION

Fracture studies of structural elements have been revolutionized in the last 50 years by the analysis of their sensitivity to flaws or cracklike defects. Within these studies an essential ingredient is reasonable and proper stress analysis especially with regard to flaws with high local elevations of stresses from which fractures progress through various crack propagation mechanisms, including corrosion and fatigue cracking.

Full studies of fracture behavior cover both the stress analysis aspects and the material behavior in terms of resistance to the stresses imposed. However, the purpose here is limited to the development of significant stress analysis details and relevant parameters, and to the compilation of available stress analysis results with cracks present insofar as they may be foreseeably related to actual fracture studies.

The redistribution of stress in a body caused by introducing a crack or notch may be solved by methods of linear-elastic stress analysis. Of course the greatest attention should be paid to the high elevation of stresses at or surrounding the crack-tip, which will usually be accompanied by at least some plasticity and other nonlinear effects. Nevertheless linear-elastic stress analysis properly forms the basis of most current fracture analysis, at least for "small scale yielding" where all substantial nonlinearity is confined within a linear-elastic field surrounding the crack-tip. Consequently, the character and significant parameters of linear-elastic crack-tip fields are examined first.

CRACK-TIP STRESS FIELDS FOR LINEAR-ELASTIC BODIES

The surfaces of a crack are the dominating influence on the distribution of stresses near and around the crack-tip, as they are the nearby and stress-free boundaries of the body. Other remote boundaries and loading forces affect only the intensity of the local stress field at the tip.

The stress fields near crack-tips can be divided into three basic types, each associated with a local mode of deformation as illustrated in **Fig. 1**. The opening mode, Mode I, is associated with local displacement in which the crack surfaces move directly apart (symmetric with respect to the $x - y$ and $x - z$ planes). The edge-sliding mode, Mode II, is characterized by displacements in which the crack surfaces slide over one another perpendicular to the leading edge of the crack (symmetric with respect to the $x - y$ plane and skew-symmetric with respect to the $x - z$ plane). Mode III, the tearing mode, finds the crack surfaces sliding with respect to one another parallel to the leading edge (skew-symmetric with respect to the $x - y$ and $x - z$ planes). The superposition of these three modes is sufficient to describe the most general three-dimensional case of local crack-tip deformation and stress fields.

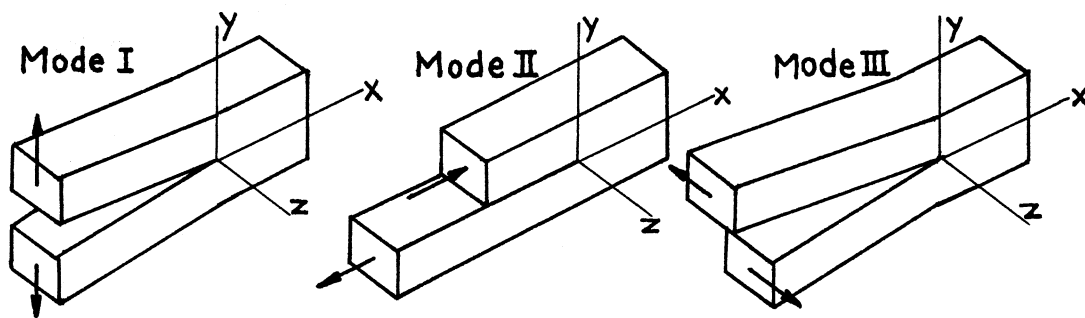


Fig. 1. Basic modes of crack surface displacements.

The most direct approach to determine the stress and displacement fields associated with each mode follows the manner of **Irwin (1957)**, which is based on the method of **Westergaard (1939)**. Modes I and II can be analyzed as two-dimensional plane-extensional problems of the theory of elasticity, which are subdivided as symmetric and skew-symmetric, respectively, with respect to the crack plane. Mode III can be regarded as the two-dimensional pure shear (or torsion) problem. Referring to **Fig. 2** for notation, the resulting stress and displacement fields are given below.

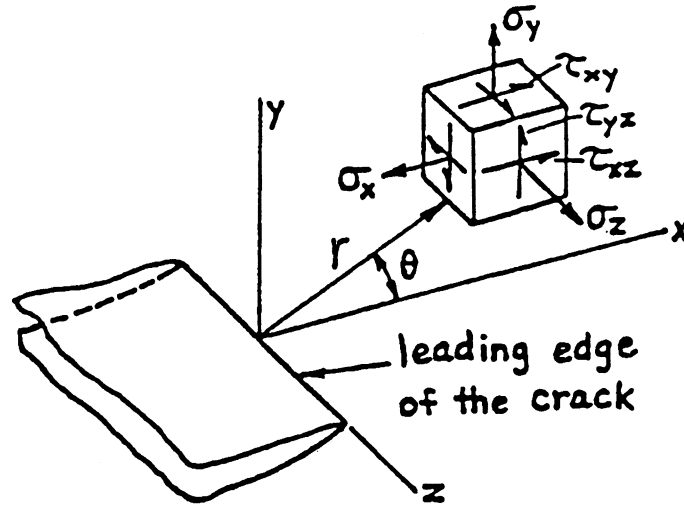


Fig. 2. Coordinates measured from leading edge of a crack and stress components in the crack-tip stress field.

Mode I:

$$\left. \begin{aligned} \sigma_x &= \frac{K_I}{(2\pi r)^{1/2}} \cos \frac{\theta}{2} \left[1 - \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \right] + \sigma_{x0} + O(r^{1/2}) \\ \sigma_y &= \frac{K_I}{(2\pi r)^{1/2}} \cos \frac{\theta}{2} \left[1 + \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \right] + O(r^{1/2}) \\ \tau_{xy} &= \frac{K_I}{(2\pi r)^{1/2}} \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cos \frac{3\theta}{2} + O(r^{1/2}) \end{aligned} \right\} \quad (1)$$

and for plane strain (with higher-order terms omitted)

$$\left. \begin{aligned} \sigma_z &= \nu(\sigma_x + \sigma_y), \quad \tau_{xz} = \tau_{yz} = 0 \\ u &= \frac{K_I}{G} [r/(2\pi)]^{1/2} \cos \frac{\theta}{2} \left[1 - 2\nu + \sin^2 \frac{\theta}{2} \right] \\ v &= \frac{K_I}{G} [r/(2\pi)]^{1/2} \sin \frac{\theta}{2} \left[2 - 2\nu - \cos^2 \frac{\theta}{2} \right] \\ w &= 0 \end{aligned} \right\}$$

Mode II:

$$\left. \begin{aligned}
 \sigma_x &= -\frac{K_{II}}{(2\pi r)^{1/2}} \sin \frac{\theta}{2} \left[2 + \cos \frac{\theta}{2} \cos \frac{3\theta}{2} \right] + \sigma_{x0} + O\left(r^{1/2}\right) \\
 \sigma_y &= \frac{K_{II}}{(2\pi r)^{1/2}} \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cos \frac{3\theta}{2} + O\left(r^{1/2}\right) \\
 \tau_{xy} &= \frac{K_{II}}{(2\pi r)^{1/2}} \cos \frac{\theta}{2} \left[1 - \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \right] + O\left(r^{1/2}\right) \\
 \text{and for plane strain (with higher-order terms omitted)} \\
 \sigma_z &= \nu(\sigma_x + \sigma_y), \tau_{xz} = \tau_{yz} = 0 \\
 u &= \frac{K_{II}}{G} [r/(2\pi)]^{1/2} \sin \frac{\theta}{2} \left[2 - 2\nu + \cos^2 \frac{\theta}{2} \right] \\
 v &= \frac{K_{II}}{G} [r/(2\pi)]^{1/2} \cos \frac{\theta}{2} \left[-1 + 2\nu + \sin^2 \frac{\theta}{2} \right] \\
 w &= 0
 \end{aligned} \right\} \quad (2)$$

Mode III:

$$\left. \begin{aligned}
 \tau_{xz} &= -\frac{K_{III}}{(2\pi r)^{1/2}} \sin \frac{\theta}{2} + \tau_{xz0} + O\left(r^{1/2}\right) \\
 \tau_{yz} &= \frac{K_{III}}{(2\pi r)^{1/2}} \cos \frac{\theta}{2} + O\left(r^{1/2}\right) \\
 \sigma_x &= \sigma_y = \sigma_z = \tau_{xy} = 0 \\
 w &= \frac{K_{III}}{G} [(2r)/\pi]^{1/2} \sin \frac{\theta}{2} \\
 u &= v = 0
 \end{aligned} \right\} \quad (3)$$

Equations (1) and (2) have been written for the case of plane strain (i.e., $w = 0$) but can be changed to plane stress easily by taking $\sigma_z = 0$ and replacing Poisson's ratio, ν , in the displacements with $\nu/(1 + \nu)$.

In **Eqs. (1) – (3)**, higher-order terms such as uniform stresses parallel to cracks, σ_{x0} , and τ_{xz0} , and terms of the order of square root of r , $O(r^{1/2})$, are as indicated. However, normally these terms are omitted since as r becomes small compared with planar dimensions (in the $x - y$ plane) of significance to the stress analysis, these higher-order terms become negligible compared with the leading $1/\sqrt{r}$ term. Therefore these leading terms are the linear-elastic crack-tip stress (and displacement) fields.

The parameters K_I , K_{II} , and K_{III} in these equations are called crack-tip stress (field) intensity factors for the corresponding three modes (**Fig. 1**). Since K_I , K_{II} , and K_{III} are not functions of the coordinates, r and θ , they represent the strength of the stress fields surrounding the crack-tip, as in **Eqs. (1)–(3)**. Alternately, they may be mathematically viewed as the strengths of the $1/\sqrt{r}$ stress singularities at the crack-tip. Their values are determined by the other boundaries of the body and the loads imposed, consequently formulas for their evaluation come from a complete stress analysis of a given configuration and loading.

Physically, K_I , K_{II} , and K_{III} may be regarded as the intensity of load transmittal through the crack-tip region caused by the introduction of a crack into the body of interest. Correspondingly, formulas for K may be regarded as formulas reflecting the redistribution of load paths for transmitting force past a crack. Thus it is plausible to observe that small amounts of plasticity or other nonlinearity at the crack-tip do not seriously further disturb the load redistribution, hence the relevance of K_I , K_{II} , and K_{III} remain.

Similarly, from a physical standpoint, K_I , K_{II} , and K_{III} may be regarded as representing the intensity of the linear-elastic stress distribution surrounding a crack-tip, where small ammounts of nonlinearity at the crack-tip

are embedded well within the field and do not significantly disturb it. Thus, a given combination of values of K_I , K_{II} , and K_{III} represents a unique crack-tip stress field environment for small-scale yielding. Because fracture processes of a material may be regarded as "caused" by this surrounding crack-tip stress field environment, the intensity factors K_I , K_{II} , and K_{III} play a large role as fracture correlation parameters in current practice. For this reason, much of the tabulated material to follow includes formulas for K_I , K_{II} , and K_{III} for various configurations and loadings.

Finally, from **Eqs. (1) – (3)** it is significant to note that stress intensity factors have units of

$$(\text{force}) \times (\text{length})^{-3/2}$$

Moreover, since they are linear factors in linear-elastic stress equations, they must be proportional to the applied loads. Thus it can be observed on a dimensional basis that in addition to the load they must contain other characteristic lengths, such as crack size. This result is a main feature of implying flaw-size effects in fracture, which indeed are observed, and further implies that these size effects can be fully analyzed only if stress analysis effects are included.

NOTE: It is interesting to note that the expressions in the brackets of the displacements u and v in **Eqs. (1)** are identical, that is,

$$\begin{aligned} 1 - 2\nu + \sin^2 \frac{\theta}{2} &\equiv 2 - 2\nu - \cos^2 \frac{\theta}{2} \\ &= \beta - \cos^2 \frac{\theta}{2} \quad (\text{see } \mathbf{P. 1.3b} \text{ and } \mathbf{P. 1.3c}) \end{aligned}$$

However, since these distinct expressions have been almost invariably used up to present, they are retained in **Eqs. (1)**. It immediately follows from this identity that the magnitude and direction of the Mode I crack-tip displacement vector $\vec{u} = (u, v)$ are given by

$$\begin{aligned} |\vec{u}| &= \frac{K_I}{G} \sqrt{\frac{r}{2\pi}} \left(\beta - \cos^2 \frac{\theta}{2} \right) \\ \frac{v}{u} &= \tan \frac{\theta}{2} \end{aligned}$$

The proportion and direction of the displacement vector in Mode I crack-tip field are schematically presented in **Fig. 3**.

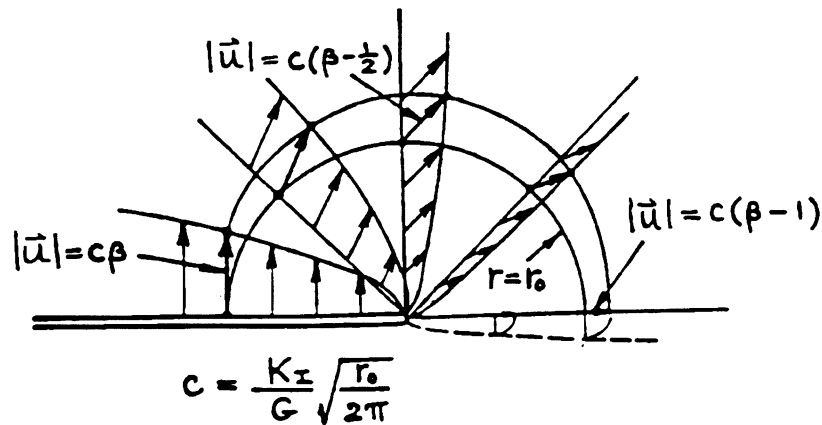


Fig. 3. Mode I displacement field.

ALTERNATE EXPRESSIONS FOR CRACK-TIP ELASTIC FIELD

In Modes I and II, stress and displacement components given by **Eqs. (1) and (2)** are sometimes expressed in alternate forms. These expressions and the corresponding expressions for $r - \theta$ components are given below.

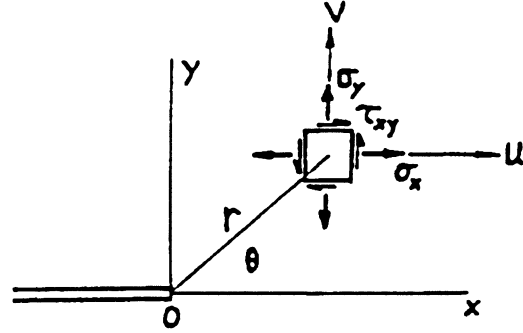


Fig. 4A

(1) $x - y$ Components (Fig. 4A)

Mode I:

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \frac{K_I}{\sqrt{2\pi r}} \cos \frac{\theta}{2} \begin{Bmatrix} 1 - \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \\ 1 + \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \\ \sin \frac{\theta}{2} \cos \frac{3\theta}{2} \end{Bmatrix} = \frac{K_I}{\sqrt{2\pi r}} \frac{1}{4} \begin{Bmatrix} 3 \cos \frac{\theta}{2} + \cos \frac{5\theta}{2} \\ 5 \cos \frac{\theta}{2} - \cos \frac{5\theta}{2} \\ -\sin \frac{\theta}{2} + \sin \frac{5\theta}{2} \end{Bmatrix}$$

$$\begin{Bmatrix} u \\ v \end{Bmatrix} = \frac{K_I}{G} \sqrt{\frac{r}{2\pi}} \begin{Bmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{Bmatrix} \left(\beta - \cos^2 \frac{\theta}{2} \right) = \frac{K_I}{G} \sqrt{\frac{r}{2\pi}} \frac{1}{4} \begin{Bmatrix} (4\beta - 3) \cos \frac{\theta}{2} - \cos \frac{3\theta}{2} \\ (4\beta - 1) \sin \frac{\theta}{2} - \sin \frac{3\theta}{2} \end{Bmatrix}$$

Mode II:

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \frac{K_{II}}{\sqrt{2\pi r}} \begin{Bmatrix} -\sin \frac{\theta}{2} \left(2 + \cos \frac{\theta}{2} \cos \frac{3\theta}{2} \right) \\ \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cos \frac{3\theta}{2} \\ \cos \frac{\theta}{2} \left(1 - \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \right) \end{Bmatrix} = \frac{K_{II}}{\sqrt{2\pi r}} \frac{1}{4} \begin{Bmatrix} -5 \sin \frac{\theta}{2} - \sin \frac{5\theta}{2} \\ -\sin \frac{\theta}{2} + \sin \frac{5\theta}{2} \\ 3 \cos \frac{\theta}{2} + \cos \frac{5\theta}{2} \end{Bmatrix}$$

$$\begin{Bmatrix} u \\ v \end{Bmatrix} = \frac{K_{II}}{G} \sqrt{\frac{r}{2\pi}} \begin{Bmatrix} \sin \frac{\theta}{2} \left(\beta + \cos^2 \frac{\theta}{2} \right) \\ -\cos \frac{\theta}{2} \left(\beta - 2 + \cos^2 \frac{\theta}{2} \right) \end{Bmatrix} = \frac{K_{II}}{G} \sqrt{\frac{r}{2\pi}} \frac{1}{4} \begin{Bmatrix} (4\beta + 1) \sin \frac{\theta}{2} + \sin \frac{3\theta}{2} \\ -(4\beta - 5) \cos \frac{\theta}{2} - \cos \frac{3\theta}{2} \end{Bmatrix}$$

where $\beta (= 1/\alpha) = \begin{cases} 2(1 - \nu) & \text{plane strain} \\ 2\left(\frac{1}{1 + \nu}\right) & \text{plain stress} \end{cases}$

(2) $r - \theta$ Components (Fig. 4B)

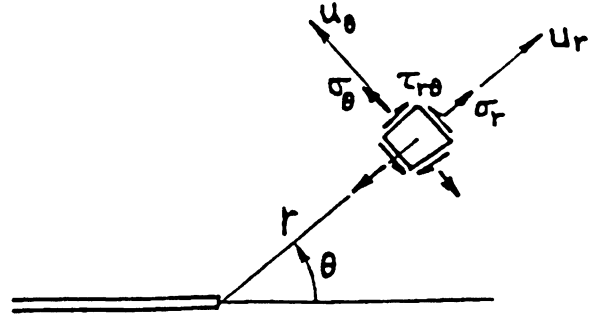


Fig. 4B

Mode I:

$$\begin{Bmatrix} \sigma_r \\ \sigma_\theta \\ \tau_{r\theta} \end{Bmatrix} = \frac{K_I}{\sqrt{2\pi r}} \cos \frac{\theta}{2} \begin{Bmatrix} 1 + \sin^2 \frac{\theta}{2} \\ \cos^2 \frac{\theta}{2} \\ \sin \frac{\theta}{2} \cos \frac{\theta}{2} \end{Bmatrix} = \frac{K_I}{\sqrt{2\pi r}} \cdot \frac{1}{4} \begin{Bmatrix} 5 \cos \frac{\theta}{2} - \cos \frac{3\theta}{2} \\ 3 \cos \frac{\theta}{2} + \cos \frac{3\theta}{2} \\ \sin \frac{\theta}{2} + \sin \frac{3\theta}{2} \end{Bmatrix}$$

$$\begin{Bmatrix} u_r \\ u_\theta \end{Bmatrix} = \frac{K_I}{G} \sqrt{\frac{r}{2\pi}} \begin{Bmatrix} \cos \frac{\theta}{2} \\ -\sin \frac{\theta}{2} \end{Bmatrix} \left(\beta - \cos^2 \frac{\theta}{2} \right) = \frac{K_I}{G} \sqrt{\frac{r}{2\pi}} \frac{1}{4} \begin{Bmatrix} (4\beta - 3) \cos \frac{\theta}{2} - \cos \frac{3\theta}{2} \\ -(4\beta - 1) \sin \frac{\theta}{2} + \sin \frac{3\theta}{2} \end{Bmatrix}$$

Mode II:

$$\begin{Bmatrix} \sigma_r \\ \sigma_\theta \\ \tau_{r\theta} \end{Bmatrix} = \frac{K_{II}}{\sqrt{2\pi r}} \begin{Bmatrix} \sin \frac{\theta}{2} \left(1 - 3 \sin^2 \frac{\theta}{2} \right) \\ -\sin \frac{\theta}{2} \left(3 \cos^2 \frac{\theta}{2} \right) \\ \cos \frac{\theta}{2} \left(1 - 3 \sin^2 \frac{\theta}{2} \right) \end{Bmatrix} = \frac{K_{II}}{\sqrt{2\pi r}} \cdot \frac{1}{4} \begin{Bmatrix} -5 \sin \frac{\theta}{2} + 3 \sin \frac{3\theta}{2} \\ -3 \sin \frac{\theta}{2} - 3 \sin \frac{3\theta}{2} \\ \cos \frac{\theta}{2} + 3 \cos \frac{3\theta}{2} \end{Bmatrix}$$

$$\begin{Bmatrix} u_r \\ u_\theta \end{Bmatrix} = \frac{K_{II}}{G} \sqrt{\frac{r}{2\pi}} \begin{Bmatrix} -\sin \frac{\theta}{2} \left(\beta - 3 \cos^2 \frac{\theta}{2} \right) \\ -\cos \frac{\theta}{2} \left(\beta + 2 - 3 \cos^2 \frac{\theta}{2} \right) \end{Bmatrix} = \frac{K_{II}}{G} \sqrt{\frac{r}{2\pi}} \frac{1}{4} \begin{Bmatrix} -(4\beta - 3) \sin \frac{\theta}{2} + 3 \sin \frac{3\theta}{2} \\ -(4\beta - 1) \cos \frac{\theta}{2} + 3 \cos \frac{3\theta}{2} \end{Bmatrix}$$

$$\text{where } \beta \left(= 1/\alpha \right) = \begin{cases} 2(1 - \nu) & \text{plane strain} \\ 2 \left(\frac{1}{1 + \nu} \right) & \text{plain stress} \end{cases}$$

NOTE: For Mode I, $u_r = u$, $u_\theta = -v$ (displacement is in $\theta/2$ direction).

SLENDER NOTCHES AND STRESS CONCENTRATIONS FROM STRESS INTENSITY FACTORS

It is worthy to note that crack-tip stress intensity factors, as detailed in **Eqs. (1)–(3)** of the previous section, are fully applicable to the tips of deep slender notches (**Creager 1967**). See **Fig. 5** for location of coordinates. For the region of the notch tip where r' is small compared with other planar ($x - y$ plane) dimensions (except for notch breadth), the stress field becomes ($\rho/2 \leq r' < \rho$, small θ')

Mode I:

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \frac{K_I}{\sqrt{2\pi r'}} \cdot \frac{\rho}{2r'} \begin{Bmatrix} -\cos \frac{3\theta'}{2} \\ \cos \frac{3\theta'}{2} \\ -\sin \frac{3\theta'}{2} \end{Bmatrix} + \frac{K_I}{\sqrt{2\pi r'}} \cos \frac{\theta'}{2} \begin{Bmatrix} 1 - \sin \frac{\theta'}{2} \sin \frac{3\theta'}{2} \\ 1 + \sin \frac{\theta'}{2} \sin \frac{3\theta'}{2} \\ \sin \frac{\theta'}{2} \cos \frac{3\theta'}{2} \end{Bmatrix} + \dots \quad (4)$$

Mode II:

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \frac{K_{II}}{\sqrt{2\pi r'}} \cdot \frac{\rho}{2r'} \begin{Bmatrix} \sin \frac{3\theta'}{2} \\ -\sin \frac{3\theta'}{2} \\ -\cos \frac{3\theta'}{2} \end{Bmatrix} + \frac{K_{II}}{\sqrt{2\pi r'}} \begin{Bmatrix} -\sin \frac{\theta'}{2} \left(2 + \cos \frac{\theta'}{2} \cos \frac{3\theta'}{2} \right) \\ \sin \frac{\theta'}{2} \cos \frac{\theta'}{2} \cos \frac{3\theta'}{2} \\ \cos \frac{\theta'}{2} \left(1 - \sin \frac{\theta'}{2} \sin \frac{3\theta'}{2} \right) \end{Bmatrix} + \dots \quad (5)$$

Mode III:

$$\begin{Bmatrix} \tau_{xz} \\ \tau_{yz} \end{Bmatrix} = \{O\} + \frac{K_{III}}{\sqrt{2\pi r'}} \begin{Bmatrix} -\sin \frac{\theta'}{2} \\ \cos \frac{\theta'}{2} \end{Bmatrix} + \dots \quad (6)$$

Note that by selecting the center of coordinates at the point $\rho/2$ from the notch tip the expansion in **Eqs. (4)–(6)** simply adds an additional term for Modes I and II when directly compared to **Eqs. (1)–(3)**. Moreover, the intensity of the added terms are also given by K_I and K_{II} and these are exactly the same K 's as found in **Eqs. (1)–(3)**. Therefore, formulas for stress intensity factors for cracks, as is extensively tabulated in this handbook, are also fully applicable to elastic stress computations for tips of slender notches.

Moreover, the first terms of **Eqs. (4) and (5)** are significant compared to the second only in the region near the end radius of the notch, $r' \rightarrow \rho/2$, whereas at greater radius—but still small compared with other planar dimensions—the same crack-tip stress field, as in **Eqs. (1) and (2)**, will dominate. Therefore, with the "disturbance" or "blunting" of a crack, or giving it a finite radius, ρ , the original crack-tip stress fields still surround the crack-tip. This is clear in the linear-elastic analysis case here, and it should be equally clear that a comparably (compared to ρ) sized zone of plasticity and/or other nonlinearity near a crack-tip is probably even less of a disturbance.

For a Mode I-type loading and configuration (i.e., K_{II} and K_{III} zero), for example, **Eq. (4)** may be used to find the stress concentration. In such a case

$$\sigma_{\max} = \sigma_y \Big|_{\substack{r = \rho/2 \\ \theta = 0}} = \frac{2K_I}{\sqrt{\pi\rho}} \quad (7)$$

This result is good for slender notches, a practical example of which is "stop-drilled" cracks (a common practice in aircraft maintenance). Applying this result to **Fig. 6**, an elliptical hole through a wide plate where

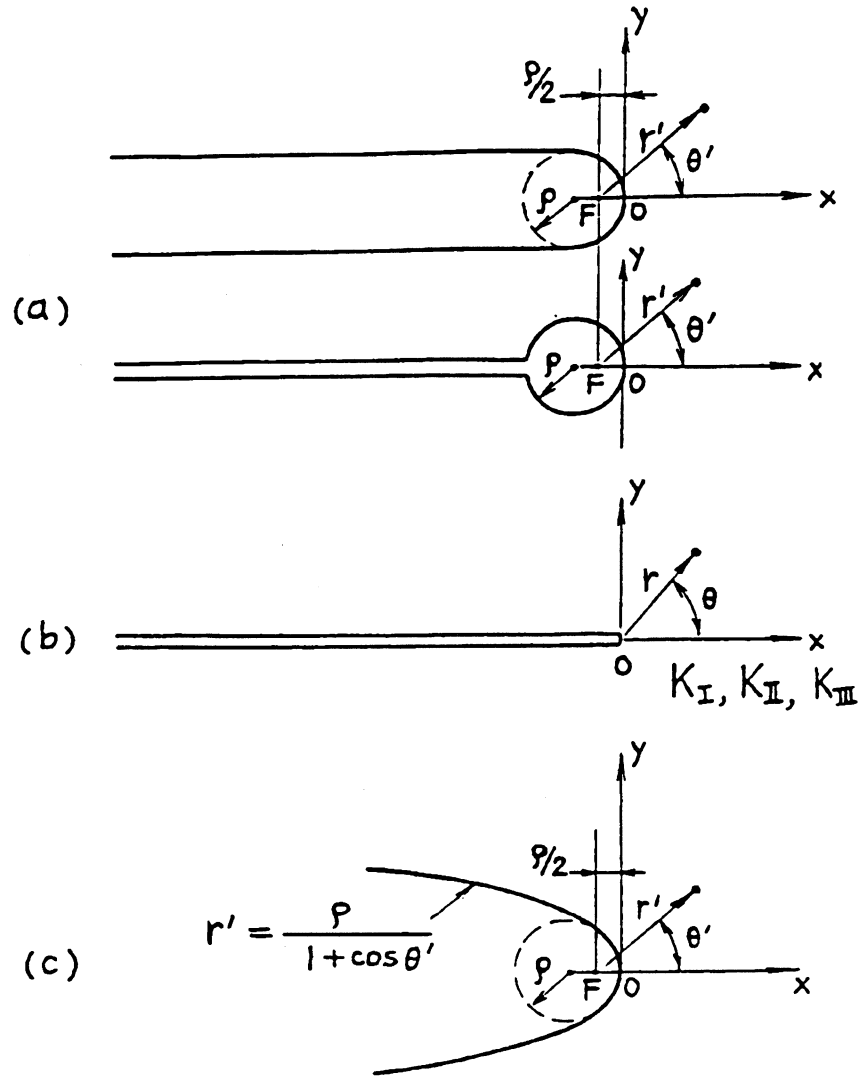


Fig. 5.

- (a) Deep slender notches and coordinate system for notch-tip stress fields.
 (b) Corresponding crack.
 (c) Conic section (parabola) used for the analysis.

the semi-major axis, a , is perpendicular to a remotely applied tension stress, σ , the comparable crack solution (see page 5.1) is

$$K_I = \sigma\sqrt{\pi a}, \quad K_{II} = K_{III} = 0 \quad (8)$$

inserting Eq. (8) into Eq. (7) gives

$$\sigma_{\max} = \sigma \left\{ 2\sqrt{\frac{a}{\rho}} \right\} \quad (9)$$

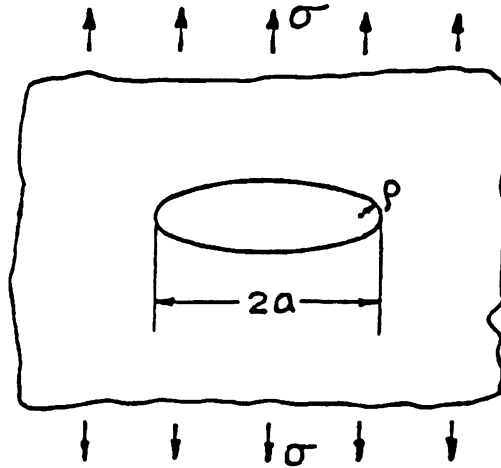


Fig. 6

The result is the well-known stress concentration solution for biaxial tension, σ , for which the comparable K_I solution is also as in **Eq. (8)**. For uniaxial stress, σ , the stress concentration result is

$$\sigma_{\max} = \sigma \left\{ 1 + 2\sqrt{\frac{a}{\rho}} \right\} \quad (10)$$

For slender notches ($\rho \ll a$), **Eqs. (9) and (10)** are in fact reasonably in agreement. The added term in **Eq. (10)** of one times σ is to be added only in cases of remotely applied uniform uniaxial stress [and is simply comparable to accounting for the additional σ_{x0} term in **Eqs. (1) and (4)**]. Knowing that **Eq. (10)** then follows from **Eq. (9)**, the full stress concentration solution followed from **Eqs. (7) and (8)**. Noting now that these stress concentration solutions, both **Eqs. (9) and (10)**, are, actually not limited to slender ellipses, the power and accuracy of this method is demonstrated.

For further demonstration, for the same elliptical hole in a large plate but loaded by equal and opposite forces (per unit plate thickness), P , on the surface of the hole at the ends of the semi-minor axis, b (see **Fig. 7**), the comparable crack solution (see **page 5.9**) is

$$K_I = \frac{P}{\sqrt{\pi a}} \quad , \quad K_{II} = K_{III} = 0 \quad (11)$$

Combining **Eqs. (7) and (11)** and noting that $\rho a = b^2$ for any ellipse gives

$$\sigma_{\max} = \frac{2P}{\pi b} \quad (12)$$

which again is the complete stress concentration solution, not limited to slender notches (**Savin 1961**).

It is not the intended purpose to present extensive information on stress concentrations here, but simply to illustrate the power of crack stress analysis. Nevertheless, it is evident that close relationships exist between stress concentration analysis and crack analysis. Later, converse to the preceding discussion, it will be pointed out that K formulas can also be derived from stress concentration formulas. For further study in stress concentration theory, **Savin (1961)**, **Neuber (1937)** and **Peterson (1953)** are recommended as starting points.

In other instances, the role and formulas for crack-tip stress intensity factors K_I , K_{II} , and K_{III} are also preserved; for example, see **Appendix D** for effects of elastic anisotropy. Hence, the most important point of this discussion of notches is to emphasize the generality and scope of crack-tip stress field analysis.

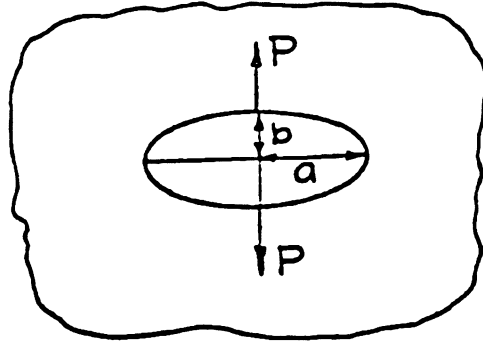


Fig. 7

ENERGY RATE ANALYSIS OF CRACK EXTENSION

Energy rate analysis of the effects of flaws historically preceded crack-tip stress field analysis. The Griffith Theory (**Griffith 1920**) and later modifications (**Irwin 1948, 1952; Orowan 1949**), termed the Griffith–Irwin Theory, made use of this approach. Basically, these methods use an energy balance analysis of crack extension.

The total elastic energy made available per unit increase in crack surface area (one side of the crack surface) is denoted by $\mathcal{G}$ for the linear-elastic case (**Irwin 1957**) (the non-linear counterpart, J , is discussed later; see **Appendices A and J**). Physically, $\mathcal{G}$, may be viewed as the energy made available for the crack extension processes at the crack-tip as a result of the work from displacements of loading forces and/or reductions in strain energy in a body accompanying a unit increase in crack area. Alternatively, $\mathcal{G}$ can be regarded as a "generalized force" based on the potential energy change per unit forward displacement of a unit length of crack front, which results in $\mathcal{G}$ being defined as "the crack extension force."

Following this line of argument, it is not difficult to show that for linear-elastic conditions (**Irwin 1948, 1952; Paris 1957**)

$$\mathcal{G} = - \frac{\partial U_T(\Delta_i, A)}{\partial A} \quad (13)$$

$$\text{and} \quad \mathcal{G} = + \frac{\partial U_T(P_i, A)}{\partial A} \quad (14)$$

$$\text{and} \quad \mathcal{G} = \frac{P^2}{2} \cdot \frac{\partial C}{\partial A} \quad (15)$$

where U_T is the total strain energy in a cracked body with a crack area A . U_T is alternately expressed in terms of A and load point displacements, Δ_i , or in terms of A and loads, P_i . In **Eq. (15)**, C is the elastic compliance and the equation is written for a single loading force, but may be generalized for several forces (**Paris 1957**). Derivations of results such as **Eqs. (13), (14), and (15)** are also available for distributed boundary tractions, and so on (**Bueckner 1958**). **Equation (15)** and its consequences are also discussed in **Appendix A**.

The $\mathcal{G}$ implied by **Eqs. (13)–(15)** is the average value along a crack front weighted for the extent of crack extension involved for each increment of crack front in the three-dimensional sense. In two-dimensional situations, such as uniform extension of a straight-through crack in a thin plate subject to extension, $\mathcal{G}$ may be

viewed as the value of a point quantity along the crack front. Moreover, for certain purposes, the three-dimensional situation may be viewed as being made up of two-dimensional slices to view $\mathcal{G}$ as a point quantity.

The expressions for $\mathcal{G}$, **Eqs. (13)–(15)**, are often useful as tools to compute $\mathcal{G}$, itself, and other quantities. For an example, see **Appendix B**, where a method of computing displacements is developed using **Eq. (14)**. **Equation (13)** will be used as an example as follows.

RELATIONSHIPS BETWEEN $\mathcal{G}$ AND K

If a cracked body is put into a "system-isolated condition," that is, with load point displacements fixed so that no work is done by loading forces, then **Eq. (13)** becomes self-evident. The energy made available for crack extension is the strain energy released by the extension.

Consider that a body with a crack is put into a system-isolated condition for conceptual clarity.

Subsequently, presume the crack-tip is elastically pulled closed over a distance α , as illustrated in **Fig. 8**, from (a) to (b). The work done in elastic closure will all go into increasing the total strain energy U_T . Therefore (**Irwin 1957; Paris 1965**)

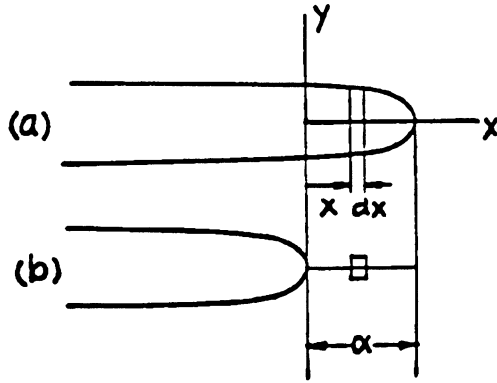


Fig. 8

$$\begin{aligned} \mathcal{G} &= - \left. \frac{\Delta U_T}{\Delta A} \right|_{\text{system isolated}} \\ &= \lim_{\alpha \rightarrow 0} \frac{2}{\alpha} \int_0^\alpha \left(\frac{\sigma_y v}{2} + \frac{\tau_{yx} u}{2} + \frac{\tau_{yz} w}{2} \right) dx \end{aligned} \quad (16)$$

Where σ_y , τ_{yx} , τ_{yz} , and u , v , w are stresses and displacements of the crack surface, respectively, occurring on the portion of the crack surface pulled closed. With the limit $\alpha \rightarrow 0$, the stress and displacements may be obtained from the crack-tip fields, **Eqs. (1)–(3)**. The stresses are appropriately obtained with $r = x$, $\theta = 0$, and the displacements

with $r = \alpha - x$, $\theta = \pi$. Making these substitutions and integrating **Eq. (16)** leads to (for plane strain)

$$\mathcal{G} = \frac{1-\nu}{2G} K_I^2 + \frac{1-\nu}{2G} K_{II}^2 + \frac{1}{2G} K_{III}^2 \quad (17)$$

It is noted that each term in the integrand of **Eq. (16)** leads to a corresponding term in **Eq. (17)** with no interaction between Modes I, II, and III. Noting $E = 2(1 + \nu)G$ and that

$$\begin{aligned} E' &= E \quad (\text{plane stress}) \\ \text{or} \\ E' &= E / (1 - \nu^2) \quad (\text{plane strain}) \end{aligned} \quad (18)$$

The total energy rate, $\mathcal{G}$, may be in general subdivided for each mode

$$\mathcal{G} = \mathcal{G}_I + \mathcal{G}_{II} + \mathcal{G}_{III} \quad (19)$$

where

$$\left. \begin{aligned} \mathcal{G}_I &= K_I^2 / E' \\ \mathcal{G}_{II} &= K_{II}^2 / E' \\ \mathcal{G}_{III} &= K_{III}^2 / 2G = \frac{1+\nu}{E} K_{III}^2 \end{aligned} \right\} \quad (20)$$

It should be noted that these relationships are for straight-ahead crack extension, as developed from **Eq. (16)** and **Fig. 8**. Therefore, although they may not suit all physical applications, they form a useful conceptual basis and are suitable for many computations.

Corresponding relationships for anisotropic elastic bodies are noted in **Appendix D**.

SUPERPOSITION OF $\mathcal{G}$ AND K RESULTS

Often in applications, a single cracked body or member is subject to several loading force systems (each system in equilibrium) which can be denoted as systems (1), (2), (3), (4), ..., etc. Because K values are factors containing the load linearly in linear-elastic stresses, superposition applies. That is to say, the total K for all loading systems applied simultaneously is the algebraic sum of K values for each system applied separately. However, because different fields of stress occur for each mode, as noted from **Eqs. (1)–(3)**, the sums must be separated for each mode or

$$\left. \begin{aligned} K_I &= K_{I(1)} + K_{I(2)} + K_{I(3)} + \dots \\ K_{II} &= K_{II(1)} + K_{II(2)} + K_{II(3)} + \dots \\ K_{III} &= K_{III(1)} + K_{III(2)} + K_{III(3)} + \dots \end{aligned} \right\} \quad (21)$$

Using **Eq. (20)** to restate **Eq. (21)** in terms of energy rates

$$\left. \begin{aligned} \mathcal{G}_I &= \left[\mathcal{G}_{I(1)}^{1/2} + \mathcal{G}_{I(2)}^{1/2} + \mathcal{G}_{I(3)}^{1/2} + \dots \right]^2 \\ \mathcal{G}_{II} &= \left[\mathcal{G}_{II(1)}^{1/2} + \mathcal{G}_{II(2)}^{1/2} + \mathcal{G}_{II(3)}^{1/2} + \dots \right]^2 \\ \mathcal{G}_{III} &= \left[\mathcal{G}_{III(1)}^{1/2} + \mathcal{G}_{III(2)}^{1/2} + \mathcal{G}_{III(3)}^{1/2} + \dots \right]^2 \end{aligned} \right\} \quad (22)$$

Equations (21) and (22) along with **Eq. (19)** become the rules for superposition of crack-tip stress intensity factors and energy rates.

It is of interest to note that **Eqs. (21)** imply that Green's function methods may be used for distributed force systems, again separating modes; although here **Eqs. (21)** are stated for discrete systems. The Green's function methods are discussed later (see **pages 1.17 and 1.18**).

MEANING OF PLANE STRESS AND PLANE STRAIN FOR FRACTURE MECHANICS PURPOSES

The mathematical definition of plane strain is that throughout a deformed body

$$\left. \begin{aligned} u &= u(x, y) \\ v &= v(x, y) \\ w &= 0(\text{or constant}) \end{aligned} \right\} \quad (23)$$

An alternate definition of plane strain is

$$\left. \begin{aligned} \varepsilon_z &= \frac{\partial w}{\partial z} = 0 \\ \gamma_{xz} &= \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} = 0 \\ \gamma_{yz} &= \frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} = 0 \end{aligned} \right\} \quad (24)$$

It can be noted that **Eqs. (24)** follow directly from **Eqs. (23)** or vice-versa. Noting Hooke's laws, yet another definition of plane strain is

$$\left. \begin{aligned} \sigma_z &= \nu(\sigma_x + \sigma_y) \\ \tau_{xz} &= 0 \\ \tau_{yz} &= 0 \end{aligned} \right\} \quad (25)$$

Again **Eqs. (25)** follow from **Eqs. (24)**.

On the other hand, plane stress is mathematically defined as

$$\left. \begin{aligned} \sigma_z &= 0 \\ \tau_{xz} &= 0 \\ \tau_{yz} &= 0 \end{aligned} \right\} \quad (26)$$

where, as before, an alternative definition in terms of strains or displacement derivatives is possible but is not of useful clarity.

The term "generalized plane stress" is applied to cases of deformation where **Eqs. (26)** apply on the average through the thickness of a thin plate subject to extensional forces. Often, when the term "plane stress" is used, "generalized plane stress" is actually implied.

The above definitions of plane stress and plane strain are those used in books and reports on the theory of elasticity, the theory of plasticity, and other such works on solid mechanics in general. However, in fracture mechanics terminology, these terms take on special, more restricted meanings. In fracture mechanics, instead of characterizing stress and strain states throughout a body, special attention is given to the crack-tip and surrounding region. "Plane stress fracture" or "plane strain fracture" have come to mean that the stress and strain conditions within the plastic zone at the crack-tip are plane stress or plane strain, respectively.

Due to the high stress-strain gradients near a crack-tip, the zone of plasticity at the tip is constrained against contraction along the crack front by the elastic material surrounding it, if the plastic zone size is small compared with the length of the crack front. This creates plane strain fracture in the "linear-elastic fracture mechanics" sense, that is, where so-called "small-scale yielding" (compared with $x - y$ planar dimensions)

conditions apply. In applications of current linear-elastic fracture mechanics analysis to flat plate test specimens (or structural members) with through-the-thickness cracks, the ratio of the crack-tip plastic zone size to sheet thickness becomes the criterion of plane stress vs. plane strain conditions. Ironically then, the most common applications of fracture mechanics stress analysis to plates with through cracks are situations where stress analysis of elastic portions of the body is properly done using plane stress to characterize conditions, but where frequently the conditions within the crack-tip plastic zone are indeed plane strain. This is called a "plane strain fracture" situation.

In applying **Eqs. (20)** to convert stress intensity factors to energy rates, or vice-versa, along with **Eq. (18)**, a confusion (if not a paradox) arises. If elastic portions of the body are plane stress and the crack-tip region is plane strain, which conditions should be used in the conversion?

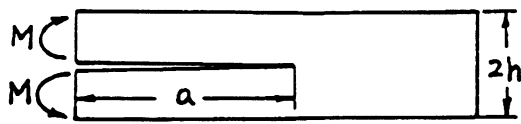


Fig. 9

For example consider a long, slender double cantilever beam configuration of constant thickness, as illustrated in **Fig. 9**. Considering the case of pure moments, M , is instructive (**Rice, 1964**), since, clearly, adding crack length is directly equivalent to adding material at the center of the cantilever arms. Plane stress conditions are thus present where all of the strain energy is added at the center of the arms. Nevertheless, conditions near the crack-tip can be plane strain, and considering the crack closure derivation of **Eqs. (20)** from **Eq. (16)**, it is also clear

that crack-tip stress field conditions apply to **Eqs. (20)** and the conversion should be made using E' for plane strain, **Eq. (18)**. Plasticity at the crack-tip tends to confound this discussion, but not if the surrounding elastic field is plane strain as well as the plastic zone. The complexity of these concepts is due to the attempt to resolve a 3-dimensional situation with 2-dimensional viewpoints, which apparently is unavoidable. Although this example illustrates that E' should be adjusted for crack-tip conditions, for simplicity, it is common practice to use the plane stress value, E , in **Eqs. (20)**. Since ν is normally about 0.3, the resulting error, if any, is less than 5% in computing K formulas. This practice is again mentioned in **Appendix A**, as compliance calibrations require conversions from $\mathcal{G}$ to K .

Finally, for elastic-plastic fracture mechanics, where small-scale yielding at a crack-tip does not apply, plane strain fracture events or tests require only that plane strain conditions exist in the "fracture process zone" at the crack-tip. This fracture process zone is the region in which the immediate crack extension processes such as advanced separations, void growth, and coalescence are taking place. This process zone may be embedded well within the crack-tip plastic zone, near the tip of the crack. Therefore it is evident that for plane strain in the process zone, it is not necessary for the whole plastic zone to be in a state of plane strain. Nevertheless, it is also evident that the immediate region of the crack-tip, that is, the process zone, must be subjected to plane strain. Therefore, blunting of a crack upon loading to a radius (or "crack opening stretch") characterized by J/σ_Y would at least require J/σ_Y to be small compared with specimen thickness, especially for through-the-thickness cracks. (Further discussions of J appear in **Appendix J**.) For this context, J may be viewed as the elastic-plastic analog to $\mathcal{G}$.

In summary, the reader is cautioned that for fracture mechanics purposes, the terms plane stress and plane strain are often used as local definitions of conditions in the crack-tip region. Moreover, the local size of the region involved will depend on the analysis approach used, linear-elastic vs. elastic-plastic, and is consequently subject to future developments in fracture analysis. Nevertheless, even within the most developed context of linear-elastic fracture mechanics confusion still remains, for the example cited of converting $\mathcal{G}$ to K , and other circumstances.

EFFECTS OF SMALL-SCALE YIELDING ON LINEAR-ELASTIC FRACTURE MECHANICS

"Small-scale yielding" means that the nonlinear zone at a crack-tip is small compared with the region in which the elastic crack-tip stress fields, **Eqs. (1)–(3)**, apply. Indeed, the circumstances are that the nonlinear or plastic zone may be regarded as embedded well within a surrounding elastic region. How small the plastic zone must be, compared to other (planar) dimensions, depends on the accuracy desired.

The size of the crack-tip plastic zone may be estimated from **Eq. (1)**, for Mode I, if small-scale yielding is in fact present. In any event, from dimensional considerations of **Eq. (1)**, it is evident that the form for an index of the size, r_Y , of the plastic zone is

$$r_Y = \alpha \left(\frac{K_I}{\sigma_{YP}} \right)^2 \quad (27)$$

where σ_{YP} is a yield strength for the material. The values of α may also be estimated from **Eq. (1)** by taking the stresses ahead of the crack, that is, $\theta = 0$, and computing the point at which yield criterion is first satisfied approaching the crack-tip (**Paris 1957**) and also adjusting for shape of the yield zone, and so on, the most commonly assumed values are (**Irwin 1960b**)

$$\alpha = \begin{cases} \frac{1}{2\pi} & (\text{plane stress}) \\ \frac{1}{6\pi} & (\text{plane strain}) \end{cases} \quad (28)$$

The plasticity at a crack-tip causes some redistribution of stresses to maintain equilibrium and therefore the full width of the plastic zone, r_P , is estimated at just twice the above results, that is

$$r_P = 2r_Y \quad (29)$$

and it is emphasized that these results, although dimensionally correct, are merely estimates, since work-hardening, large strains, and other obvious influences are ignored.

The redistribution of stress to satisfy equilibrium implies that the center of coordinates (r, θ) for the elastic field, **Eqs. (1)**, is advanced ahead of the real crack-tip into the zone of plasticity (**Paris 1957; Irwin 1960b**). This correction for the "effective crack size" is often taken as approximately equal to r_Y added to the actual crack size. Using an "effective crack size" in linear-elastic fracture mechanics stress analysis calculation is regarded as sometimes appropriate. However, if very high accuracy is desired, it is appropriate to have r_Y small enough compared to planar dimensions, including crack size, that it may be entirely neglected. On the other hand, for the purpose of examining trends, and for low-accuracy calculations, the "effective crack size" correction has been proposed at times for application to large-scale yield situations. In any event, judgment is required for particular applications.

Nevertheless, provided the scale of yielding is small enough, all of the preceding results and derivations based on linear-elastic theory are, indeed, correct and appropriate to apply to real physical situations. Therefore, this discussion now proceeds based mainly on linear-elastic analysis and methods, but it will also provide other nonlinear analyses and mathematical models as seems appropriate for possible future use (e.g., see **Appendix J**).

INTRODUCTION TO STRESS FUNCTION METHODS

As shown by the preceding discussion, the primary objective of crack stress field analysis is to obtain a characterization of the stress-strain region enclosing the crack-tip, the region within which the progressive separational process occurs. Characterization in terms of K values, assuming linear-elastic behavior, only requires knowledge of stresses and strains close to the crack-tip. However, studies of crack extension often involve displacement measurements at some distance from the crack-tip. Thus solutions of crack problems that permit stress and displacement calculations in the entire stress field are of interest. Solutions of Mode I and Mode II crack problems in closed form are known for a large number of 2-dimensional, linear-elastic problems. The solution procedure uses the stress function approach and, therefore, the stress function method is discussed first. Except for special problems, mainly those of Mode III type, closed-form solutions are strictly applicable only to infinite plate crack problems. Computations of K values and displacements for strip and finite-plate crack problems usually require a numerical approach. In such problems, however, the stress function viewpoint can often enhance the efficiency of numerical methods.

Choosing x, y Cartesian coordinates, the stress equilibrium equations are

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0, \quad \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} = 0 \quad (30)$$

These equations are satisfied if we assume

$$\sigma_x = \frac{\partial^2 \Phi}{\partial y^2}, \quad \sigma_y = \frac{\partial^2 \Phi}{\partial x^2}, \quad \tau_{xy} = -\frac{\partial^2 \Phi}{\partial x \partial y} \quad (31)$$

The Hooke's Law equations are

$$\left. \begin{aligned} E\varepsilon_x &= \sigma_x - \nu(\sigma_y + \sigma_z) \\ E\varepsilon_y &= \sigma_y - \nu(\sigma_x + \sigma_z) \\ E\gamma_{xy} &= 2(1 + \nu)\tau_{xy} \end{aligned} \right\} \quad (32)$$

where $\sigma_z = 0$ for plane stress and $\sigma_z = \nu(\sigma_x + \sigma_y)$ for plane strain and the identity $E = 2G(1 + \nu)$ can be noted.

A convenient equation representing the fact that three strains are defined in terms of derivatives of only two displacements (strain compatibility) is given by

$$\frac{\partial^2 \varepsilon_x}{\partial y^2} + \frac{\partial^2 \varepsilon_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y} \quad (33)$$

Substituting **Eq. (31)** into **Eq. (32)**, followed by use of **Eq. (33)**, provides

$$\begin{aligned} \nabla^2 (\nabla^2 \Phi) &= 0 \\ \text{where } \nabla^2 &= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \end{aligned} \quad (34)$$

Equation (34) is obtained independently of whether plane-stress or plane-strain is assumed in **Eq. (32)**; Φ is termed the Airy stress function.

As a starting point for solving specific problems, **Muskhelishvili (1933)** noted certain analysis advantages were possible if one assumed the solution of **Eq. (34)** was either the real or the imaginary part of

$$F = z^* \phi(z) + \chi(z) \quad (35)$$

where $z = x + iy$ and $z^* = x - iy$. If the problem can be arranged so that the crack of interest occupies a straight segment of the x -axis ($y = 0$), a simpler, one-function approach suggested by **Westergaard (1939)** is often useful. Westergaard discussed several Mode I crack problems that could be solved using

$$\Phi = \operatorname{Re}\{\bar{Z}(z)\} + y \operatorname{Im}\{\bar{Z}(z)\} \quad (36)$$

where

$$\left. \begin{aligned} \bar{Z} &= \frac{d}{dz} \bar{Z} \\ Z &= \frac{d}{dz} Z, \quad Z' = \frac{d}{dz} Z \end{aligned} \right\} \quad (37)$$

From **Eqs. (36) and (31)**

$$\left. \begin{aligned} \sigma_x &= \operatorname{Re} Z - y \operatorname{Im} Z' \\ \sigma_y &= \operatorname{Re} Z + y \operatorname{Im} Z' \\ \tau_{xy} &= -y \operatorname{Re} Z' \end{aligned} \right\} \quad (38)$$

For a straight crack on $y = 0$, a loading symmetry such that $\tau_{xy} = 0$ on $y = 0$, corresponding to Mode I, is automatically furnished by **Eq. (36)**. The displacements, assuming plane-strain, are given by

$$\left. \begin{aligned} 2Gu &= (1 - 2\nu) \operatorname{Re} \bar{Z} - y \operatorname{Im} Z \\ 2Gv &= 2(1 - \nu) \operatorname{Im} \bar{Z} - y \operatorname{Re} Z \end{aligned} \right\} \quad (39)$$

For plane-stress, ν in **Eq. (39)** can be replaced by $\nu/(1 + \nu)$.

In checking the derivation of **Eq. (38)** from **Eq. (36)** and of **Eq. (39)** from integration of **Eq. (32)**, it is helpful to use the Cauchy-Riemann equations. These are

$$\left. \begin{aligned} \operatorname{Re}\{f'(z)\} &= \frac{\partial}{\partial x}(\operatorname{Re} f) = \frac{\partial}{\partial y}(\operatorname{Im} f) \\ \operatorname{Im}\{f'(z)\} &= \frac{\partial}{\partial x}(\operatorname{Im} f) = -\frac{\partial}{\partial y}(\operatorname{Re} f) \end{aligned} \right\} \quad (40)$$

In terms of **Eq. (35)**, **Eq. (36)** corresponds to assuming

$$\Phi = \operatorname{Re} F, \quad \chi = \bar{Z} - z\phi, \quad \phi = \frac{1}{2} \bar{Z} \quad (41)$$

The solution of one of the crack problems briefly discussed in **Westergaard (1939)** is given by

$$Z(z) = \frac{\sigma}{\sqrt{1 - (a/z)^2}} \quad (42)$$

The problem solved with this stress function is the crack problem studied by **Griffith (1920)** with the aid of previous work by **Inglis (1913)**; a central crack of length $2a$, with $\sigma_y = \sigma_x = \sigma$ at distances remote from the crack.

In terms of the vectors

$$z = re^{i\theta}, \quad z - a = r_1 e^{i\theta_1}, \quad z + a = r_2 e^{i\theta_2} \quad (43)$$

Equation (42) can be expressed as

$$Z = \frac{\sigma r}{\sqrt{r_1 r_2}} e^{i\left\{\theta - \frac{\theta_1 + \theta_2}{2}\right\}} \quad (44)$$

From differentiation of **Eq. (42)**

$$Z'(z) = \frac{-\sigma a^2}{\left\{z^2 - a^2\right\}^{3/2}} \quad (45)$$

and can be expressed as

$$Z' = \frac{-\sigma a^2}{(r_1 r_2)^{3/2}} e^{-i\frac{3}{2}(\theta_1 + \theta_2)} \quad (46)$$

From integration of **Eq. (42)**,

$$\bar{Z}(z) = \sigma \sqrt{z^2 - a^2} \quad (47)$$

and can be expressed as

$$\bar{Z} = \sigma \sqrt{r_1 r_2} e^{i\left(\frac{\theta_1 + \theta_2}{2}\right)} \quad (48)$$

The angles in the preceding equations are restricted to the range $-\pi$ to π (radians). **Equations (44), (46), and (48)** are helpful in forming the real and imaginary parts of functions as indicated in the equations for stresses and displacements using the identity $e^{i\phi} = \cos \phi + i \sin \phi$. From these equations it is clear that $\text{Re} Z, y \text{ Re} Z'$,

and $y \operatorname{Im} Z'$ are all zero along the line segment occupied by the crack $|x| < a$ and $y = 0$. Thus free boundary conditions along lines of the crack are provided. Remote from the crack, as $|z|$ approaches infinity, $y \operatorname{Re} Z'$ and $y \operatorname{Im} Z'$ are again zero and $\operatorname{Re} Z = \sigma$. Thus the remote stress field is $\sigma_y = \sigma_x = \sigma$ and $\tau_{xy} = 0$.

In the limit of small enough values of r_1/a , taking $r = a$, $r_2 = 2a$, $\theta = 0$, and $\theta_2 = 0$, **Eq. (44)** becomes

$$Z = \frac{\sigma \sqrt{a}}{\sqrt{2r_1}} e^{-i\frac{\theta_1}{2}} \quad (49)$$

This relation can be written as

$$Z(\zeta) = K / \sqrt{2\pi\zeta} \quad (50)$$

$$\text{where } \zeta = r_1 e^{i\theta_1} = z - a \quad (51)$$

$$\text{and } K = K_I = \sigma \sqrt{\pi a} \quad (52)$$

The Mode I stresses and displacements very close to the crack-tip (as shown in the introductory comments) can be derived using **Eq. (50)**, the associated values of Z' and $\bar{Z}$, expressing these in vector form as illustrated above, and substituting real and imaginary parts (as appropriate) into **Eqs. (38) and (39)**.

The single stress function approach of Westergaard is conveniently extended to Mode II crack problems by assuming (**Irwin 1958a**)

$$\Phi = -y \operatorname{Re} \bar{Z} \quad (53)$$

In terms of **Eq. (35)**, **Eq. (53)** corresponds to the choices

$$\Phi = \operatorname{Im} F, \quad \chi = -z\phi, \quad \phi = \frac{1}{2} \bar{Z} \quad (54)$$

The stresses are given by

$$\left. \begin{aligned} \sigma_x &= 2 \operatorname{Im} Z + y \operatorname{Re} Z' \\ \sigma_y &= -y \operatorname{Re} Z' \\ \tau_{xy} &= \operatorname{Re} Z - y \operatorname{Im} Z' \end{aligned} \right\} \quad (55)$$

The displacements (plane strain) are given by

$$\left. \begin{aligned} 2Gu &= 2(1 - \nu) \operatorname{Im} \bar{Z} + y \operatorname{Re} Z \\ 2Gv &= -(1 - 2\nu) \operatorname{Re} \bar{Z} - y \operatorname{Im} Z \end{aligned} \right\} \quad (56)$$

The solution of the Mode II counterpart of the Griffith crack problem is obtained by

$$Z(z) = \frac{\tau}{\sqrt{1 - (a/z)^2}} \quad (57)$$

The remote stresses are $\sigma_x = \sigma_y = 0$, $\tau_{xy} = \tau$. The crack-tip stresses and displacements are again provided by **Eq. (50)** with

$$K = K_{II} = \tau\sqrt{\pi a} \quad (58)$$

The use of essentially the same stress function, Z , to solve Mode I and Mode II problems is applicable to many crack stress field problems and can be extended to Mode III by means of the equation

$$Gw = \text{Im}\bar{Z} \quad (59)$$

The stresses are given by

$$\left. \begin{aligned} \tau_{yz} &= \text{Re}Z \\ \tau_{xz} &= \text{Im}Z \end{aligned} \right\} \quad (60)$$

Further use of a Z function, which solves a two-dimensional Mode I crack problem in an isotropic material, in the solution of two-dimensional crack problems (of similar configuration) in orthotropic and anisotropic elastic materials is discussed in **Appendix D**. To provide relations that remain generally valid, it is most convenient to define the three K values as follows [consistent with definitions in **Eqs.(1)–(3)**]:

$$\left\{ \begin{array}{c} K_I \\ K_{II} \\ K_{III} \end{array} \right\} = \lim_{r \rightarrow 0} \sqrt{2\pi r} \left\{ \begin{array}{c} \sigma_y \\ \tau_{xy} \\ \tau_{yz} \end{array} \right\} \quad (61)$$

where r is the length of a small vector extending directly forward from the crack-tip.

In the case of Mode I, the invariants used in computing principal stresses are

$$\frac{\sigma_x + \sigma_y}{2} = \text{Re}Z, \quad \tau_{\max} = y|Z'| \quad (62)$$

For plane strain, the stress field energy density, U , is given by

$$U = \frac{1 - 2\nu}{2G} (\text{Re}Z)^2 + \frac{1}{2G} \tau_{\max}^2 \quad (63)$$

The corresponding relationships for Mode II are

$$\frac{\sigma_x + \sigma_y}{2} = \text{Im}Z, \quad \tau_{\max} = \sqrt{|Z|^2 + y^2 |Z'|^2 - 2y \text{Im}(Z^* Z')} \quad (64)$$

$$U = \frac{1 - 2\nu}{2G} (\text{Im}Z)^2 + \frac{1}{2G} \tau_{\max}^2 \quad (65)$$

Williams (1957) called attention to the possibility that studies of U near the leading edge of a crack might be of interest in predicting crack extension behavior. **Irwin (1958a)** noted that the largest tensile stresses at a fixed small distance from a Mode I crack-tip were at 60° to the line of expected crack extension. Either viewpoint predicts a tendency for the location of advance separation to cause roughening of a flat tensile fracture surface. Of course, the subject of these comments pertains to the fracture process zone and a treatment based on stress-strain relations within the crack-tip plastic zone would be more appropriate.

ADDITIVITY OF CRACK STRESS FIELDS AND K VALUES

From the additivity of linear-elastic stress fields and the definitions of K , **Eq. (61)**, several conclusions are evident: (a) the addition of a stress field that does not possess an inverse square root stress singularity at the crack-tip does not alter the value of K for that crack-tip; (b) when each of several superimposed stress fields contributes to the K values, the K values are separately additive for each of these modes; and (c) when several loading configurations are applied to the same crack and the Westergaard Z functions for each are known, the Z functions can be added together, and the stresses and displacements can then be derived from the total Z function using methods discussed in the previous section.

For illustration consider the Z function

$$Z(z) = \frac{P}{\pi(z-b)} \cdot \frac{\sqrt{a^2 - b^2}}{\sqrt{z^2 - a^2}} \quad (66)$$

Using the Mode I value of Φ , **Eq. (36)**, the problem solved is that of a central crack of length, $2a$, opened by a pair of splitting forces, P , acting against the crack surfaces at the position $y = 0$, $x = b$. The value of K at $x = a$ is

$$K = \frac{P}{\sqrt{\pi a}} \sqrt{\frac{a+b}{a-b}} \quad (67)$$

If we add a second pair of equal size splitting forces at $y = 0$, $x = -b$, the total value of Z becomes

$$Z(z) = \frac{2P}{\pi(z^2 - b^2)} \frac{\sqrt{a^2 - b^2}}{\sqrt{1 - (a/z)^2}} \quad (68)$$

From **Eq. (68)**, the K value at each crack-tip is given by

$$K = \frac{2P}{\sqrt{\pi a}} \cdot \frac{a}{\sqrt{a^2 - b^2}} \quad (69)$$

Simple addition shows that **Eq. (69)** is the sum of **Eq. (67)** plus the same expression after substitution of $-b$ for b inside the radical. Assume next that $P = \sigma db$. From the additivity rule, the stress field for a uniform pressure, σ , acting against the crack surfaces can be derived from the following Z function:

$$Z(z) = \frac{2\sigma}{\pi\sqrt{1 - (a/z)^2}} \int_0^a \frac{\sqrt{a^2 - b^2} db}{z^2 - b^2} \quad (70)$$

Performing the integration,

$$Z(z) = \frac{\sigma}{\sqrt{1 - (a/z)^2}} - \sigma \quad (71)$$

Eq. (71) could have been derived by the alternative method of adding a uniform biaxial compression, $Z = -\sigma$, to the Z value for the Griffith crack (**Eq. 42**). Because the uniform stress field does not alter K , the value of K at each crack-tip is $\sigma\sqrt{\pi a}$, as in **Eq. (52)**. If the result needed is the total K rather than the total Z , a substantial simplification of the computational task can be expected. For example, elementary methods show that

$$\int_0^a \frac{db}{\sqrt{a^2 - b^2}} = \frac{\pi}{2} \quad (72)$$

Use of $P = \sigma db$ in **Eq. (69)** and use of **Eq. (72)** provides $\sigma\sqrt{\pi a}$. The simplicity of this computation can be compared to the integration indicated in **Eq. (70)**.

From the additivity principles just illustrated, it can be seen that the solution of a crack stress field problem can be visualized as a two-step process: (1) solve the stress distribution problem in a manner satisfying the boundary conditions (including applied loads) but with the crack considered absent; (2) add to this stress field a stress field that cancels any stresses acting directly across the crack along the line of the crack. In the case of a crack occupying a segment of the x -axis, the stresses along this segment which must be reduced to zero are σ_y , τ_{xy} , and τ_{yz} . Closed-form solutions of numerous infinite-plate crack problems have been obtained in this way. Because of the analysis simplifications applicable to Mode III, closed-form solutions can be obtained using this method for certain finite plate problems. The two-step approach can be termed a Green's function method when a suitable stress function for local pressures or shears on the crack surfaces is available. A suitable stress function of the Green's function type is one that can be added to the "no crack" stress field without inconsistency with the boundary conditions assumed in the first step of the above method.

BOUNDARY COLLOCATION METHOD

The availability of large, high-speed computers permits a variety of numerical methods that can be used when K cannot be found from a closed-form crack stress field solution. Boundary collocation can be regarded as a relatively simple extension of methods discussed previously.

Assume that the crack occupies a segment of the x -axis with the crack-tip at $z = 0$, and that both of the loads and the shape of the plate containing the crack are symmetrical relative to the x -axis. A simple example would be a long, single-edge-notched tensile specimen with a crack-simulating notch of length, a , open to the left free boundary of the plate, and with uniform tension, σ , applied across the upper and lower boundaries of the plate (parallel to the crack). Let W be the width of the plate and let L be the length. The stress field is of Mode I type and consideration can be given to the use of the stress function Z , where

$$Z(z) = \frac{K}{\sqrt{2\pi z}} + \sum_{n=1}^N A_n z^{n-1/2} \quad (73)$$

Using **Eq. (38)**, σ_y and τ_{xy} are zero on $y = 0$ when x is negative. Thus free boundary conditions are exactly satisfied along the line of the crack. In addition, it can be observed that $E\nu = 2 \operatorname{Im}(\bar{Z}) = 0$ on $y = 0$ for positive values of x . Since it is desirable to restrict N to a moderate size, free boundary conditions cannot be exactly satisfied along $x = -a$ and $x = W - a$. However, if the values of $A_1, A_2, \dots, A_N$, and K are such that free boundary conditions are nearly satisfied along these lines, the influence of the remaining errors on the

stress field close to the crack will be small. A similar consideration applies to the upper and lower boundaries, where the desired boundary conditions, $\sigma_y = \sigma$ and $\tau_{xy} = 0$, can be satisfied only on an average basis. Along the lines $x = -a$ and $x = W - a$, since stress field errors more remote from the crack-tip are of lesser importance, it is convenient to choose the boundary collocation points at y values corresponding to equal spacing of u_1 in the equation $y = a \tan u_1$, where u_1 is an angle measured from the negative branch of the x -axis. A similar method for choosing boundary collocation points on $x = W - a$ might be to use $y = (W - a) \tan u_2$, where u_2 is measured from the positive branch of the x -axis. The preceding methods are continued as a means of selecting boundary collocation points across the line $y = L/2$. Only the specimen half above the x -axis is used because of the symmetry of the problem.

The solution procedure consists of writing the equations for $\tau_{xy} = 0$ and $\sigma_x = 0$ at the points selected along $x = -a$ and $x = W - a$ as well as the equations for $\tau_{xy} = 0$ and $\sigma_y = 1$ (since K is proportional to σ) at points selected along $y = L/2$. If $\frac{1}{2}(N + 1)$ boundary points are selected, the result is a set of equations, linear in terms of the parameters $K, A_1, A_2, \dots, A_N$, and just sufficient in number to permit determination of each parameter. However, for a given amount of computing time, it has proved most efficient to limit the value of N , select the number of boundary collocation points that is three to four times $\frac{1}{2}(N + 1)$, and use a least squares program to determine the best values for the parameters. Selection of boundary collocation points at sharp corners should be avoided. The outputs needed from the computer are the value of K and (usually) the value of the y -direction displacement at the crack mouth position commonly selected for clip gage crack opening measurements during crack toughness evaluations. A number of calculation refinements can be added. However, only the basic plan of the method is presented here.

SUCCESSIVE BOUNDARY STRESS CORRECTION METHOD

For illustration, consider a straight, two-dimensional crack occupying the segment of the x -axis, $0 \leq x \leq a$. Assume that the y -axis is the free boundary of a semi-infinite plate and that the stresses remote from the crack are $\sigma_y = \sigma$, $\sigma_x = 0$, $\tau_{xy} = 0$. From previous comments, the K is not altered if the remote stresses are all taken as zero and a uniform pressure, σ , is assumed acting inside of the crack. Thus an approximate estimate of K is given by $K \simeq \sigma\sqrt{\pi a}$. An appropriate stress field is provided by the Z function of **Eq. (71)** under the assumption that σ would be added to real calculation of σ_y . Along $x = 0$, the value of τ_{xy} from this Z function is zero and the average of σ_x is zero. Successive additions of stress fields which remove the normal stresses, σ_x , along $x = 0$, and the consequent normal stresses σ_y along the line of the crack, can be visualized as an infinite repetition of additive stress fields (**Irwin 1962a**). Each stress removal along the line of the crack provides a corrective contribution to the value of K . If the σ_x stresses on $x = 0$ from the Z function of **Eq. (71)** are termed $\sigma_x(\sigma)$, the calculation can be compactly summarized by a pair of integral equations as follows:

$$p(y_0) = \sigma_x(\sigma) + \int_0^a q(b)f(b, y_0)db \quad (74)$$

$$q(b) = \int_0^\infty p(y_0)g(b, y_0)dy_0 \quad (75)$$

where

$$\sigma_x(\sigma) = \sigma \left\{ \frac{y_0 (y_0^2 + 2a^2)}{(y_0^2 + a^2)^{3/2}} - 1 \right\}$$

$$f(b, y_0) = \frac{2}{\pi} \frac{y_0 \sqrt{a^2 - b^2}}{(y_0^2 + b^2) \sqrt{a^2 + y_0^2}} \left(\frac{2y_0^2}{y_0^2 + b^2} - \frac{2a^2 + y_0^2}{a^2 + y_0^2} \right) \quad (76)$$

$$g(b, y_0) = -\frac{4}{\pi} \frac{by_0^2}{(b^2 + y_0^2)^2} \quad (77)$$

The total K is given by

$$K = \sqrt{\pi a} \left\{ \sigma + \frac{2}{\pi} \int_0^a \frac{q(b)db}{\sqrt{a^2 - b^2}} \right\} \quad (78)$$

and turns out to be larger than $\sigma\sqrt{\pi a}$ by about one-eighth. For calculation purposes it is convenient to choose $\sigma = 1$ (since K is proportional to σ), $b = a \sin \alpha$, $y_0 = a \tan \beta$, and to use equal intervals of α and β in conducting the necessary numerical integrations. The calculation is started by using of $\sigma_x(\sigma)$ for $p(y_0)$ in **Eq. (75)**. The resulting $q(b)$ is used in **Eq. (74)** to obtain an improved value of $p(y_0)$ for use in **Eq. (75)**. A calculation of K from **Eq. (78)** is made for each new value of $q(b)$ and the computing process is stopped when the change of K becomes smaller than some selected fraction of $\sigma\sqrt{\pi a}$. Similar calculation plans are applicable if the initial "no-crack" stress field is due to thermal stress induced by a uniform rapid cooling along $x = 0$ (**Lachenbruch 1961**).

The use of the preceding method for strip problems is more complex. Even for problems having symmetry relative to the midline of the strip, convenient stress removal functions for the parallel free boundaries of the strip are not available. Nevertheless, a number of problems of this type have been solved (**Tada 1972b**) using a modification of the method. The main advantage of the successive stress removal method is rapid convergence of the total K estimate. In compensation, the programing tasks may be substantially greater than would be encountered using the boundary collocation method.

K ESTIMATES FROM FINITE ELEMENT METHODS*

Although finite element methods are commonly used for practical stress analysis problems, special planning is necessary for efficient use of such methods to determine K for two-dimensional crack problems. For three-dimensional crack problems—for example, a part-through surface crack—even with expert planning, an accuracy of 10% is not easily obtained. The comments given here are mainly limited to two-dimensional problems.

Two kinds of procedures have been used with considerable success. In the first of these, assuming the stress state is caused by externally applied loads, the method is directed toward the computation of the total stress field energy, $U_T = \frac{1}{2} \sum P_i \Delta_i$, for each of a series of crack sizes with the loads, P_i , held constant. In the preceding equation, Δ_i is the displacement parallel to P_i of the local region of load application, If δA is the

* Historically, this was written for the First Edition in 1973. Progress in numerical methods continues at a rapid pace.

increase of the severed area corresponding to a crack size increment and if δU_T is the increase of U_T for that increment, K can be obtained from the equation

$$\frac{K^2}{E} = \mathcal{G} = \frac{\delta U_T}{\delta A} = \frac{1}{2} \sum P_i \frac{\delta \Delta_i}{\delta A}$$

This procedure simply models an experimental compliance calibration in terms of finite element computations of the load displacements, Δ_i (**Watwood 1969**). The advantage of the procedure is that refinement of the finite element mesh toward very small element sizes near the crack-tip has little influence on the displacement differences, $\delta \Delta_i$, remote from the crack and is therefore not required. In compensation, the computations must include a range of crack sizes and reliable results require careful study of the accuracy of the load displacement differences.

Procedures that use very small elements close to the crack-tip have yielded values of K with an accuracy of better than 2%. Experience is necessary in developing approximate methods of size reduction of the finite elements close to the crack-tip and in choosing the stiffness matrix for this region. Generally, K values have been determined by positioning computed values of an extensional stress at the element centroid; plotting values of stress times the square root of distance from the crack-tip, r , against r ; and extrapolating to $r = 0$. A discussion of finite element methods for K determination by **Kobayashi (1973)** suggests that, in terms of computational efficiency, using crack line displacements at nodal points close to the crack-tip along with the displacement equations valid for the crack-tip region possesses advantages over procedures based on the stress equations.

Currently, K estimates from finite element methods are at an intermediate stage of development. Trials of calculations of $\mathcal{G}$ in terms of the J -Integral using finite element methods may provide substantial advantages. In addition, the fact that the stress and displacement patterns at the crack-tip are known and that approximate first-order corrections to these patterns can be estimated has not been fully exploited (**Wilson 1972**). The latter method may be essential as a means for obtaining accurate K estimates for three-dimensional crack problems within reasonable limits of computational expense (**Swedlow 1972**).

ADDITIONAL REMARKS FOR PART I

A. Unified Formulation for In-Plane Two-Dimensional Problems

It is well known that all formulas for in-plane, two-dimensional problems can be expressed in common forms for plane-strain and plane-stress conditions by choosing G and one other elastic constant which is defined in terms of ν according to the condition.

The expression: $G = E/2(1 + \nu)$ can be used commonly for both plane-strain and plane-stress conditions. For the second constant, κ defined as follows is frequently used.

$$\kappa = \begin{cases} 3 - 4\nu & \text{plane strain} \\ \frac{3-\nu}{1+\nu} & \text{plane stress} \end{cases} \quad (79)$$

A choice of a neater combination, however, is $(1 - \nu)$ for plane strain and $1/(1 + \nu)$ for plane stress. Two constants, α and β ($\beta = 1/\alpha$), are defined in this handbook as follows for convenience:

$$\beta = \frac{1}{\alpha} = \begin{cases} 2(1 - \nu) & \text{plane strain} \\ 2/(1 + \nu) & \text{plane stress} \end{cases} \quad (80)$$

A complete, unified formulation in terms of stresses in the x,y Cartesian coordinate system, for example, is summarized as follows.

$$\left. \begin{array}{l}
 \text{Equilibrium equations:} \\
 \text{Compatibility equation:} \\
 \text{Stress-Strain relations:} \\
 \text{where}
 \end{array} \right\} \begin{array}{l}
 \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + X = 0 \\
 \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + Y = 0 \\
 \nabla^2 \sigma^* = -\operatorname{div} \vec{F} = -\left(\frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y}\right) \\
 2G\varepsilon_x = \sigma^* - \sigma_y \\
 2G\varepsilon_y = \sigma^* - \sigma_x \\
 2G\gamma_{xy} = 2\tau_{xy} \\
 \sigma^* = \beta(\sigma_x + \sigma_y)/2 \\
 \vec{F}(X, Y) = \text{body force}
 \end{array} \quad (81)$$

Note that $\sigma_z = \nu(\sigma_x + \sigma_y)$ in plane-strain condition or $\varepsilon_z = -\nu(\sigma_x + \sigma_y)/E$ in plane-stress condition [see **Eqs. (25) and (26)**] is determined simply as a by-product of the analysis.

Various distinct expressions are used throughout this handbook for plane strain and plane stress. All of these pairs of expressions can be unified to single forms with the use of G and α or β . Some examples of unified expressions are

$$\text{a. Eq. (79): } \kappa = 2\beta - 1 \quad (79a)$$

$$\text{b. Eq. (39): } \left. \begin{array}{l} 2Gu = (\beta - 1) \operatorname{Re} \bar{Z} - y \operatorname{Im} Z \\ 2Gv = \beta \operatorname{Im} \bar{Z} - y \operatorname{Re} Z \end{array} \right\} \quad (39a)$$

Also refer to **Eq. (1)** and **pages 1.3b and 1.3c**.

$$\text{c. Eq. (18): } E' = 4\alpha G \quad (18a)$$

d. **Page 16.3:**

$$\begin{aligned}
 v &= \pm \frac{1 - \alpha}{G} \sigma h \\
 K_I &= 2\sqrt{\alpha(1 - \alpha)} \sigma \sqrt{h}
 \end{aligned} \quad (82)$$

Also note that $\alpha = 1/2(1 - \nu)$ repeatedly appears in solutions for many three-dimensional problems. See, for example, **pages 23.7 and 24.1**.

B. On Completeness of Westergaard Single-Function Method for Analysis of Cracks

The single stress function approach of Westergaard has a certain deficiency that only affects the elastic field in the absence of cracks. For the analysis of the contribution to the elastic field by the presence of cracks, the single-function method is complete.

The deficiency consists in the possible presence of non-zero Z_{II} even in the so-called Mode I field, for example, in the absence of cracks (**Okamura 1976**), as illustrated here in the analysis of a simple example.

The subscripts I and II are used here, as in the solution pages, to represent Mode I and Mode II fields, respectively.

A general Airy stress function Φ in terms of Westergaard stress functions is given, combining **Eqs. (36) and (53)**, by

$$\Phi = \Phi_I + \Phi_{II} = \operatorname{Re} \bar{Z}_I + y \operatorname{Im} \bar{Z}_I - y \operatorname{Re} \bar{Z}_{II} \quad (83)$$

The corresponding stresses and displacements, combining **Eqs. (38) and (55)** and **Eqs. (39) and (56)**, respectively, are given by

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \begin{Bmatrix} \operatorname{Re} Z_I - y \operatorname{Im} Z_I' \\ \operatorname{Re} Z_I + y \operatorname{Im} Z_I' \\ -y \operatorname{Re} Z_I' \end{Bmatrix} + \begin{Bmatrix} 2 \operatorname{Im} Z_{II} + y \operatorname{Re} Z_{II}' \\ -y \operatorname{Re} Z_{II}' \\ \operatorname{Re} Z_{II} - y \operatorname{Im} Z_{II}' \end{Bmatrix} \quad (84)$$

$$2G \begin{Bmatrix} u \\ v \end{Bmatrix} = \begin{Bmatrix} (\beta - 1) \operatorname{Re} \bar{Z}_I - y \operatorname{Im} \bar{Z}_I \\ \beta \operatorname{Im} \bar{Z}_I - y \operatorname{Re} \bar{Z}_I \end{Bmatrix} + \begin{Bmatrix} \beta \operatorname{Im} \bar{Z}_{II} + y \operatorname{Re} \bar{Z}_{II} \\ -(\beta - 1) \operatorname{Re} \bar{Z}_{II} - y \operatorname{Im} \bar{Z}_{II} \end{Bmatrix} \quad (85)$$

where $\bar{Z}_I(z)$, $\bar{Z}_{II}(z)$, etc., are abbreviated to $\bar{Z}_I$, $\bar{Z}_{II}$, etc.

Consider, for example, an infinite plate subjected to (a) uniform (biaxial) tension $\sigma_x = \sigma_y = \sigma$ at infinity, **Fig. 10(a)**, and (b) uniaxial tension $\sigma = \sigma$ at infinity, **Fig. 10(b)**. Both elastic fields are symmetric with respect to x -axis and therefore are Mode I fields. Nevertheless, Z_{II} for (b) is not zero, but $Z_{II} = -i(\sigma/2)$. That is, the Westergaard stress functions and, for reference, the Airy stress functions corresponding to **Fig. 10(a) and (b)** are given by

$$(a) \quad Z_I(z) = \sigma, \quad Z_{II}(z) = 0; \quad \Phi = \frac{1}{2} \sigma (x^2 + y^2) \quad (86a)$$

$$(b) \quad Z_I(z) = \sigma, \quad Z_{II}(z) = -i \frac{\sigma}{2}; \quad \Phi = \frac{1}{2} \sigma x^2 \quad (86b)$$

These Westergaard functions, **Eqs. (86a) and (86b)**, obviously from **Eq. (84)**, yield identical stresses on the x -axis (the presence of nonzero Z_{II} in (b) has no bearing on them), namely,

$$\sigma_y(x, 0) = \sigma, \quad \tau_{xy}(x, 0) = 0 \quad (87)$$

Therefore, to make a segment $|x| \leq a$, $y = 0$ traction-free surfaces, the stress field shown in **Fig. 11** must be superimposed on those of **Fig. 10**. The Westergaard function corresponding to **Fig. 11** has been obtained in the preceding section, that is, **Eq. (71)**.

$$Z_I(z) = \frac{\sigma}{\sqrt{1 - (a/z)^2}} - \sigma, \quad Z_{II}(z) = 0 \quad (71)$$

As readily observed from the analysis, as long as the crack-absent stress distributions to be removed over crack segments are identical, the contributions to the elastic field by the presence of cracks are identical regardless of the difference in the overall elastic fields, and cracks in Mode I fields contribute to Z_I only.

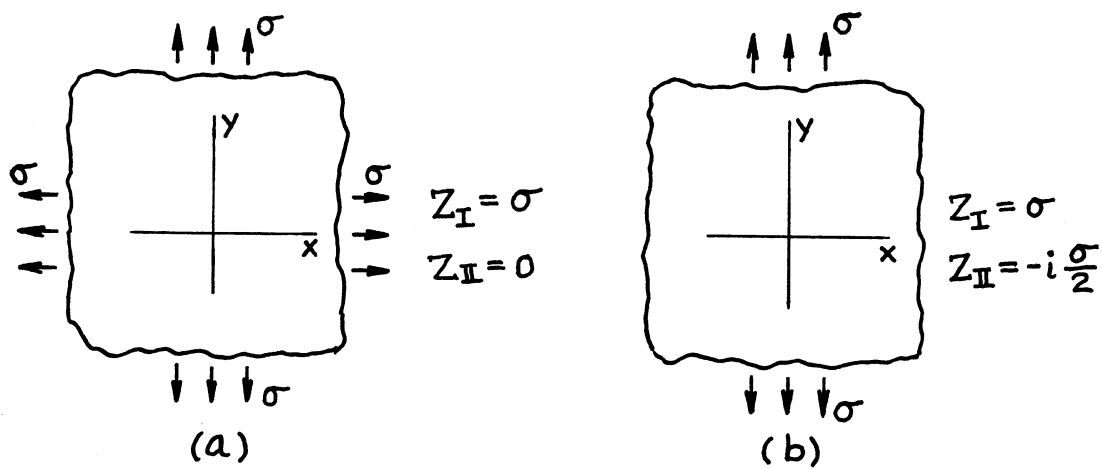


Fig. 10

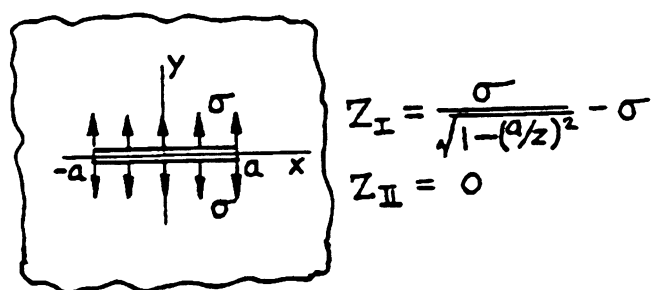


Fig. 11

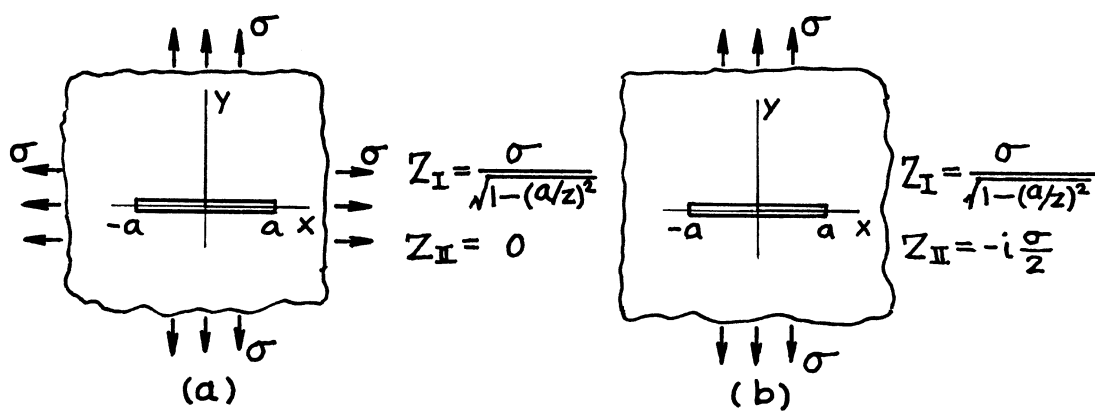


Fig. 12
(Fig. 10) + (Fig. 11)

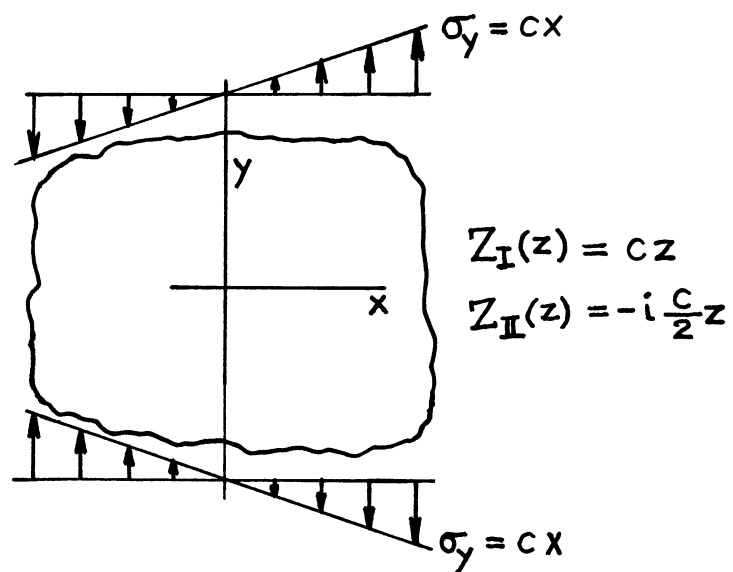


Fig. 13

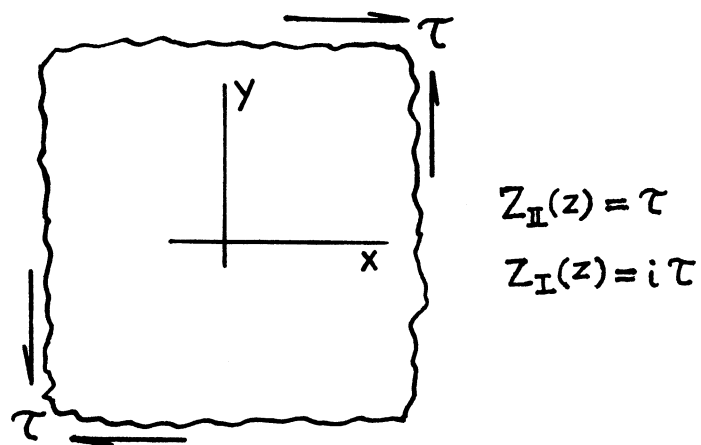


Fig. 14